Nonlinear interactions in a hybrid multi-field nanoresonator subject to combined external and parametric excitation

R. Alcorta[a], A. Gemintern[a], R. H. Blick[b], O.Gottlieb[a]

[a] *Technion - Israel Institute of Technology, Department of Mechanical Engineering. Haifa, 32000, Israel*
[b] *Universität Hamburg-Faculty of Mathematics, Informatics and Natural Science. Institute for Nanostructure and Solid State Physics. Luruper Chaussee 149 22761 Hamburg, Germany*

Abstract

With the advent of micro- and nano-electromechanical systems (MEMS/NEMS) in the past two decades, robust devices with very high frequency response have been designed, such as RF mechanical mixers. To achieve mixing, a nanoresonator must be made to oscillate in the nonlinear regime through quasi-periodic forcing. Since nonlinearities arise from a variety of sources in NEMS -e.g. large displacements, electromechanical coupling, fluid-structure interaction and boundary conditions-, constructing models that adequately capture the associated effects and allow for accurate prediction is a crucial practical problem, especially in light of the abundant experimental evidence of bifurcations and chaos in such systems. The main objective of the present work is two-fold: firstly, we systematically construct a quasi-continuum model for the dynamics of a multi-field nanoresonator and consistently incorporate both the effects of geometric nonlinearity and nonlinear contributions from magneto-motive and electrodynamic excitations. We then obtain a low-dimensional dynamical system by exploiting symmetry and using modal projection. Secondly, we numerically explore the system's response in a variety of situations: single-mode and two-mode approximations subjected to magneto-motive (periodic), electrodynamic (parametric) or combined (quasi-periodic) excitations. Both periodic solution continuation with a frequency-domain solver and classical time integration are used to generate frequency responses, bifurcation analyses, spectral densities and Poincaré maps. The results presented here showcase the richness of the proposed model, as evidenced by the diversity of phenomena encountered: modal interactions, symmetry breaking, sub-harmonic cascades, torus doubling, and chaos.

1. Introduction

Over the course of the past two decades, nanomechanical resonators combining the advantages of mechanical systems and silicon electronics have found application in numerous fields, including: high-precision mass sensing [1, 2], charge detection [3], electron shuttles [4], scanning probe microscopy [5] fundamental research at the frontier between the classical and quantum realms [6], and ultradense memory [7]. The appeal for nanoresonators in these and other contexts is due to their very high frequency response [8, 9], high quality factor and tunability [21], and silicon-based fabrication akin to that of conventional electronic circuits.

A notable application in the field of radio-frequency signal processing is mechanical mixing, wherein the usual nonlinear circuits responsible for frequency conversion are replaced by flexible elements. With respect to the fully-electronic equivalent, this approach offers the advantages of enhanced robustness to external electrical disturbances and low-energy consumption. In practice, commonly-used structural elements for mixing include nanopillars [11] or multi-field nanobeams [12]. The latter is in line with the concept of employing coupled rather than single-field resonators for increased capability. Kambali et al. [13] studied the response of a base-coupled microcantilever array to electrodynamic excitation and revealed both quasiperiodic and nonstationary energy transfers between array elements. Lyu et al [14] exploited nonlinear mode localization for high-precision mass sensing in an electrostatically-coupled

microresonator array. The resolution of a sensor consisting of a capacitively-actuated microcantilever beam coupled to a nanowire was shown to be orders of magnitude higher than the system employing the beam alone by Vidal-Alvarez et al. [1].

Actuation of nanomechnical resonators is frequently achieved through capacitive plates, which exert a force proportional to the square of a given input signal and to a nonlinear function of the instantaneous displacement fields [15]. The input, in the context of mixing, must additionally contain at least two mutually incommensurate frequencies. Consequently, the system is subjected to an excitation which is simultaneously parametric and quasiperiodic. Compared to studies on single-frequency (periodic and/or parametric) excitation, reports of quasiperiodically excited beams are rather rare. Moon and Holmes [16] performed experiments on a buckled-beam Duffing oscillator under two-frequency excitation and observed chaotic strange attractors through the so-called double Poincaré section technique. Wiggins [17] employed a Melnikov function approach to explain these experimental results analytically, thus proposing a criterion for the existence of chaos through transverse homoclinic tori. A thorough exploration of the bifurcation structure and chaotic dynamics of a quasiperiodically-forced, doubly-clamped beam was performed by Yagasaki [18]. Therein, the author proved that chaos in reduced-order models is a sufficient condition for the continuous model to exhibit similar behaviour. Tan and Kang [19] numerically studied the response of a Duffing oscillator to combined parametric and quasiperiodic excitation as a model for automotive disk brake pad instability, and observed different routes to chaos near the system's natural and parametric resonance frequencies. A single-field, clamped-clamped nanoresonator subjected to quasiperiodic capacitive excitation was recently considered by Gusso et al. [20], who used a single-mode approximation and observed robust chaos over the parameter space of interest.

To the authors' best knowledge, studies to date are lacking on models which simultaneously incorporate the following three features, fundamental for accurately predicting the dynamics of a mechanical mixer: i) a consistent derivation of the reduced-order equations of motion from a nonlinear continuum model, ii) combined quasi-periodic and parametric excitation, and iii) consideration of modal interaction effects. In the present work, we aim to address this deficiency by proposing and investigating a quasi-continuum nonlinear initial boundary value problem for the three-field nanoresonator introduced by [10]. The construction of this problem incorporates geometric nonlinearity with the nonlinear contributions of magneto-motive and electrodynamic excitations in a consistent fashion. Through the use of symmetry and modal projection, the spatio-temporal field equations are reduced to a low-order, nonlinearly-coupled system of oscillators subjected to quasi-periodic excitation. Governing parameters are estimated from available experimental measurements for small-amplitude response. The system is investigated numerically for both magneto-motive and combined magneto-motive/electrodynamic excitations, with responses analysed through power spectra and their associated Poincaré map.

This paper is organised as follows: Section 2 presents the derivation of a non-dimensional, reduced-order model for the resonator dynamics, starting from the continuous equations of motion for a nonlinear beam. Values of the physical parameters governing these equations are estimated from previous experiments in Section 3. Effective hinge stiffness and axial stress -both unavailable from direct measurements- are seen to have an important influence on the system's properties, and especially on the eigenfrequencies of the reduced model. This aspect is explored in Section 4. In Sections 5 and 7, the system's response to magneto-motive and mixed (magneto-motive plus capacitive) excitations, respectively, is numerically analysed. In both cases, results from single-mode and two-mode models are presented and contrasted via numerical continuation, power spectra and Poincaré maps. A brief discussion of the obtained results is addressed in Section 8. Finally, closing remarks are presented in Section 9.

2. Problem formulation

2.1. Initial boundary-value problem

We consider the small-amplitude dynamics of a thin three-field elastic nanoresonator with total length $2L$ and rectangular cross-section $A = bh$, as shown in Fig. 1. This configuration represents the experimental system studied by Schreibe et al. [10, 21, 22]. A coordinate system is chosen whose origin coincides with the beam's midspan, with x representing the horizontal coordinate and departure from equilibrium in the vertical direction described by the spatio-temporal field $v(x,t)$. At two points, located at $\pm L/2$ from the origin, the system is constrained to zero displacement. These connections are modelled in this work as hinges with effective torsional stiffness k, which allows for mutual excitation between the outer and inner fields. Furthermore, a thin conducting covering enables combined magnetomotive excitation at midspan -with current i in a magnetic field of strength B- and electrodynamic excitation at the outer edges through alternating voltage capacitors. The magnitude of the latter depends on the distance between the grounded beam surface -identified here with the beam centreline- and the biased edge of the corresponding capacitor, equal to g on both sides for the beam at equilibrium.

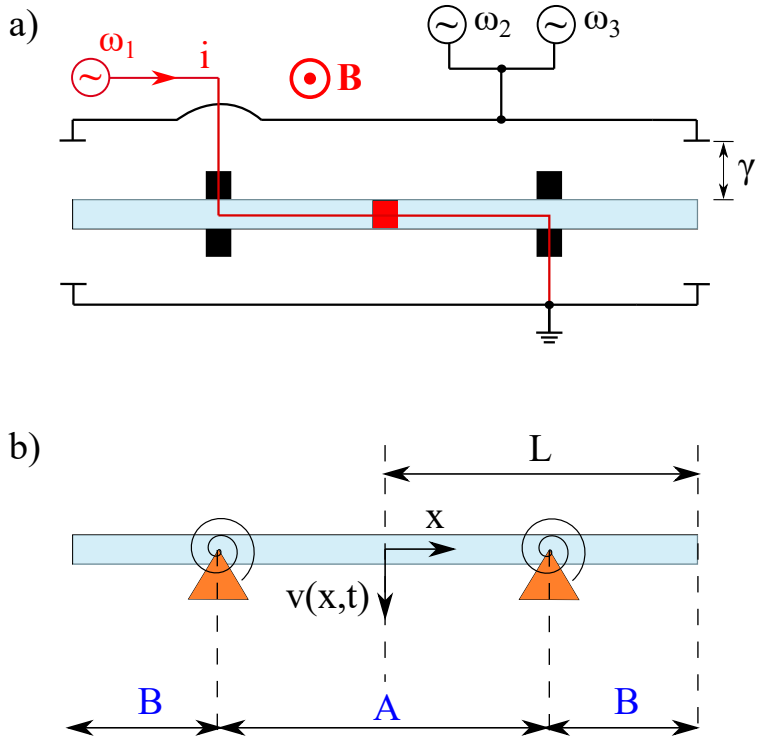


Figure 1: Definition sketch: a) representation of the experimental nanoresonator, b) reduced two-field scheme.

Two distinct displacement fields are defined for the inner and outer beams: $v^A(x,t)$, and $v^B(x,t)$, respectively. We use Euler-Bernoulli theory to describe these fields and a Kelvin-Voigt model [23] to relate stresses and strains within the beam, the latter choice being motivated by the experimental low-temperature conditions (around 4.2K [10]). This leads to viscoelastic damping terms in both fields, plus nonlinear stiffness and damping due to mid-plane stretching in the constrained field A [24]. The effect of an axial load with magnitude N, due to potential residual stresses, is likewise considered. Finally, we incorporate both a magnetomotive force [25] and edge capacitors [15] as localized

forces at $x = 0$ and $x = \pm L/2$, respectively.
This yields the following governing equations:

$$\rho A v^A_{tt} + EI v^A_{xxxx} + \left[N - \frac{EA}{2L} \int_{-\frac{L}{2}}^{\frac{L}{2}} \left(v^A_x \right)^2 dx - \frac{DA}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} v^A_x v^A_{xt} dx \right] v^A_{xx} = F - DI v^A_{xxxxt} \quad x \in [-L/2, L/2] \tag{1}$$

$$\rho A v^B_{tt} + EI v^B_{xxxx} = G - DI v^B_{xxxxt} \qquad x \in [-L, -L/2] \cup [L/2, L] \tag{2}$$

in which subscripts signify differentiation with respect to space (x) or time (t). The localized forces are given explicitly by:

$$\begin{cases} F = -\hat{\delta}(x) \left[iB \sin(\omega_1 t) \right] \left[1 - \frac{1}{2} \left(v^A_x \right)^2 \right] \\ G = \hat{\delta}(x \pm L) \frac{\epsilon_0 b}{2} \left[\frac{V_2 \sin(\omega_2 t) + V_3 \sin(\omega_3 t)}{g - v^B} \right]^2 \end{cases} \tag{3}$$

where $\hat{\delta}(\cdot)$ is the Dirac delta function, and ρA, EI and D are the mass per unit length, stiffness and damping coefficients of the beam, respectively. In addition, we find the magnetic parameters i, ω_1, and B which represent the magnitude and frequency of the alternating current and the magnetic field, respectively. The capacitor voltages and frequencies, respectively, are denoted by V_j, ω_j, with $j = 2, 3$. We shall assume, in this paper, that $V_2 = V_3 = V$.

A well-posed mathematical description of the system is completed through the addition of supplementary equations, which can be categorized into two groups: (i) zero-moment and zero-shear force boundary conditions of the free edges at $\pm L$:

$$v^B_{xx}(\pm L, t) = v^B_{xxx}(\pm L, t) = 0 \tag{4}$$

(ii) compatibility conditions at the hinges, ensuring zero displacements and continuity of slopes and moments:

$$\begin{aligned} & v^A(\pm L/2, t) = v^B(\pm L/2, t) = 0 \quad , \quad v^A_x(\pm L/2, t) = v^B_x(\pm L/2, t) \\ & EI \left[v^A_{xx}(\pm L/2, t) - v^B_{xx}(\pm L/2, t) \right] = k v^A_x(\pm L/2, t) \end{aligned} \tag{5}$$

2.2. Nondimensional formulation

Before proceeding to obtain a reduced-order model of our system, the field equations -along with boundary and compatibility conditions- are rescaled by the length (L) and an elastic frequency scale defined by: $\omega_s^2 = EI/\rho A L^4$. Hence, we define the quantities: $s = x/L$, $u = v/L$, $\tau = \omega_s t$, which are introduced in the above equations to yield a non-dimensional initial boundary-value problem for the nanoresonator:

$$u^A_{\tau\tau} + u^A_{ssss} + \left[\tilde{\beta} - \tilde{\alpha} \int_{-\frac{1}{2}}^{\frac{1}{2}} \left(u^A_s \right)^2 ds - \tilde{\eta} \int_{-\frac{1}{2}}^{\frac{1}{2}} u^A_s u^A_{s\tau} ds \right] u^A_{ss} = \tilde{F} - \tilde{\zeta} u^A_{sssst} \qquad s \in [-1/2, 1/2] \tag{6}$$

$$u^B_{\tau\tau} + u^B_{ssss} = \Gamma - \tilde{\zeta} u^B_{sssst} \qquad s \in [-1, -1/2] \cup [1/2, 1] \tag{7}$$

$$\begin{cases} \tilde{F} = -\hat{\delta}(s) \tilde{\mu} \sin\left(\tilde{\Omega}_1 \tau \right) \left[1 - \frac{1}{2} \left(u^A_s \right)^2 \right] \\ \Gamma = \hat{\delta}(s \pm 1) \tilde{\epsilon} \left[\frac{\sin\left(\tilde{\Omega}_2 \tau \right) + \sin\left(\tilde{\Omega}_3 \tau \right)}{\gamma - u^B} \right]^2 \end{cases} \tag{8}$$

with the non-dimensional boundary and compatibility conditions:

$$u^B_{ss}(\pm 1,\tau) = u^B_{sss}(\pm 1,\tau) = 0 \tag{9}$$

$$\begin{aligned} &u^A(\pm 1/2,\tau) = u^B(\pm 1/2,\tau) = 0 \quad , \quad u^A_s(\pm 1/2,\tau) = u^B_s(\pm 1/2,\tau) \\ &\left[u^A_{ss}(\pm 1/2,t) - u^B_{ss}(\pm 1/2,t)\right] = \kappa u^A_s(\pm 1/2,\tau) \end{aligned} \tag{10}$$

The system is governed by the following set of eight non-dimensional parameters:

$$\begin{aligned} &\tilde{\alpha} = \frac{AL^2}{2I} \qquad \tilde{\beta} = \frac{NL^2}{EI} \qquad \kappa = \frac{kL}{EI} \qquad \tilde{\mu} = \frac{iBL^3}{EI} \\ &\tilde{\zeta} = \frac{D}{L^2}\sqrt{\frac{I}{\rho AE}} \quad \tilde{\eta} = D\sqrt{\frac{A}{\rho EI}} \quad \gamma = \frac{g}{L} \quad \tilde{\epsilon} = \frac{\epsilon_0 V^2 bL}{2EI} \end{aligned} \tag{11}$$

Likewise, we have introduced the non-dimensional frequencies $\tilde{\Omega}_j = \omega_j/\omega_s$, $j = 1, 2, 3$.

2.3. Modal dynamical system

The full IVBP to be solved consists of two coupled fields, Eqs. (6)-(7), along with boundary and compatibility conditions, Eqs. (9) and (10), respectively. We make use of a Galerkin procedure to reduce the continuous IVBP to a discrete set of ordinary differential equations. To this end, we assume a modal decomposition for each displacement field in terms of symmetric and anti-symmetric vibration modes of the beam, such that: $u^A(s,\tau) = \sum_n q_n(\tau)\phi^A_n(s)$, $u^B(s,\tau) = \sum_n q_n(\tau)\phi^B_n(s)$. Herein, $q_n(\tau)$ are the modal displacements and the mode shape functions $\left[\phi^A_n(s), \phi^B_n(s)\right]$ are derived in Appendix A as solutions to the underlying linear, unforced and undamped problem. This requires augmenting the above system of equations by two additional continuity constraints at $s = 0$, defining either a symmetric or an anti-symmetric mode shape. Respectively, these conditions read:

$$u^A_s(0,\tau) = u^A_{sss}(0,\tau) = 0 \qquad \text{or:} \qquad u^A(0,\tau) = u^A_{ss}(0,\tau) = 0 \tag{12}$$

For each n, the global shape function $\phi_n(s) = \phi^B_n(s) \bigcup \phi^A_n(s)$ is smooth and continuous over $s \in [-1, 1]$, as seen in Fig. 2 for the first two modes of the system.

Following [10], we focus our attention on the primary resonance of the first mode. Hence, we employ a two-term modal expansion where $u^j(s,t) = \phi^j_1(s)q_1(\tau) + \phi^j_2(s)q_2(\tau)$, for $j = A, B$, wherein $q_1(\tau)$ and $q_2(\tau)$ are the amplitudes of the first symmetric and anti-symmetric modes, respectively.

Substitution of these into Eqs. (6) through (7), multiplication of the resulting equations by ϕ_j, $j = 1, 2$ and integration over the whole beam span yields the following system of coupled differential equations for the modal dynamics:

$$q_{1,\tau\tau} + \tilde{\zeta} z_1^4 q_{1,\tau} + \left[z_1^4 + \tilde{\beta} I_1\right] q_1 + \tilde{\alpha}\left[\tilde{a}_{11} q_1^3 + \tilde{a}_{12} q_1 q_2^2\right] + \tilde{\eta}\left[\tilde{a}_{11} q_1^2 q_{1,\tau} + \tilde{a}_{12} q_1 q_2 q_{2,\tau}\right] = \tilde{f}_1 + \tilde{g}_1 \tag{13}$$

$$q_{2,\tau\tau} + \tilde{\zeta} z_2^4 q_{2,\tau} + \left[z_2^4 + \tilde{\beta} I_2\right] q_2 + \tilde{\alpha}\left[\tilde{a}_{21} q_1^2 q_2 + \tilde{a}_{22} q_2^3\right] + \tilde{\eta}\left[\tilde{a}_{21} q_1 q_2 q_{1,\tau} + \tilde{a}_{22} q_2^2 q_{2,\tau}\right] = \tilde{g}_2 \tag{14}$$

where the coefficients $\tilde{a}_{11}, ..., \tilde{a}_{21}$ are obtained from the integration of mode shape function products, and are detailed in Appendix B. Moreover, the quantities z_1 and z_2 are the non-dimensional elastic wave numbers associated to the first and second modes, respectively (see Appendix A). We remark that the chosen basis leads to a diagonal linear part, which implies that we can identify these modes with the first two normal modes of the linearised system.

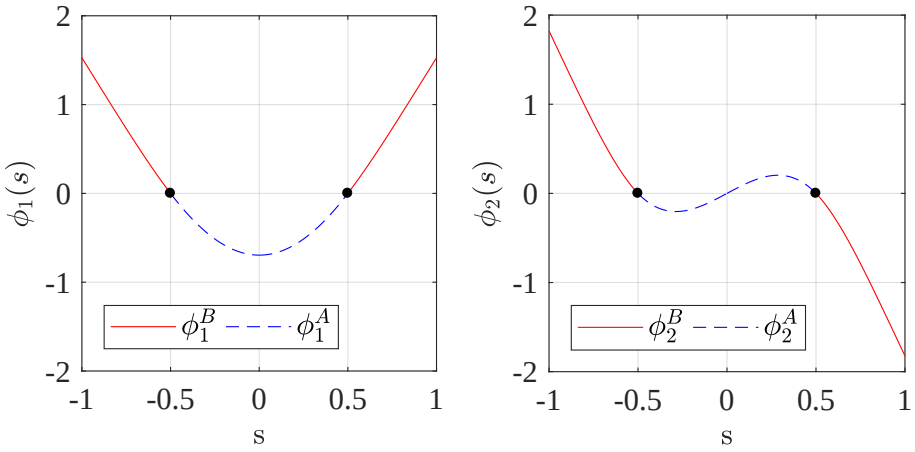


Figure 2: First symmetric (left) and anti-symmetric (right) bending modes.

Moreover, as a consequence of mode shape orthogonality, modal projection of stretching-induced effects does not produce fully-coupled equations: for instance, no term proportional to q_2^3 appears in Eq. (13). The projected external forces read:

$$\begin{cases} \tilde{f}_1 = \tilde{\mu}\left[\phi_1^A(0)\right]\sin(\tilde{\Omega}_1\tau)\left[1-\frac{1}{2}\left[\left(\phi_2^A\right)'(0)\right]^2 q_2^2\right] \\ \tilde{g}_{1,2} = \tilde{\epsilon}\left[\phi_{1,2}^B(1)\right]\left\{\left[\frac{\sin(\tilde{\Omega}_2\tau)+\sin(\tilde{\Omega}_3\tau)}{\gamma-\left(\left[\phi_1^B(1)\right]q_1+\left[\phi_2^B(1)\right]q_2\right)}\right]^2 \pm \left[\frac{\sin(\tilde{\Omega}_2\tau)+\sin(\tilde{\Omega}_3\tau)}{\gamma-\left(\left[\phi_1^B(1)\right]q_1-\left[\phi_2^B(1)\right]q_2\right)}\right]^2\right\} \end{cases} \tag{15}$$

where the positive and negative signs in the latter correspond to $\tilde{g}_1$ and $\tilde{g}_2$, respectively. We note that, as the magnetomotive force is localized on a node for anti-symmetric modes, it has no contribution to Eq. (14). Conversely, since $\left[\left(\phi_2^A\right)'(0)\right] \neq 0$, $\tilde{f}_1$ introduces a parametric excitation term into Eq. (13) which is absent from the single-mode approximation.

We simplify the modal equations further by rescaling displacements and time according to:

$$y_1 = \left[\phi_1^B(1)\right]\frac{q_1}{\gamma} \quad ; \quad y_2 = \left[\phi_2^B(1)\right]\frac{q_2}{\gamma} \quad ; \quad \hat{t} = \hat{\omega}_1\tau$$

where $\hat{\omega}_1^2 = z_1^4 + \tilde{\beta} I_1$ is the squared non-dimensional frequency of the first mode. By making use of these new variables in Eqs. (13), (14), we finally obtain the dynamical system describing rescaled modal displacements $y_1(\hat{t})$ and $y_2(\hat{t})$, as follows:

$$\ddot{y}_1 + \zeta_1\dot{y}_1 + y_1 + \alpha_{11}y_1^3 + \alpha_{12}y_1y_2^2 + \eta_{11}y_1^2\dot{y}_1 + \eta_{12}y_1y_2\dot{y}_2 = f_1 + g_1 \tag{16}$$

$$\ddot{y}_2 + \zeta_2\dot{y}_2 + r^2y_2 + \alpha_{21}y_1^2y_2 + \alpha_{22}y_2^3 + \eta_{21}y_1y_2\dot{y}_1 + \eta_{22}y_2^2\dot{y}_2 = g_2 \tag{17}$$

Here, dots represent derivatives with respect to $\hat{t}$, the linear damping coefficients are given by: $\zeta_{1,2} = \tilde{\zeta} z_{1,2}^4/\hat{\omega}_1$, and $r = \hat{\omega}_{1a}/\hat{\omega}_{1s}$ is the ratio of linear natural frequencies. The sets of nonlinear damping and curvature coefficients, respectively η_{ij} and α_{ij}, are proportional to $\tilde{\eta}$ and $\hat{\alpha}$, respectively, while also depending on mode shape coefficients ($\phi_{1,2}^B(1)$), first-mode natural frequency (ω_{1a}) and dimensionless gap size ($\tilde{\gamma}$) (Appendix C). Likewise, the external

forces simplify to the following expressions:

$$\begin{cases} f_1 = \mu \sin(\Omega_1 \hat{t}) \left[1 - \dfrac{1}{2} d_0^2 y_2^2\right] \\ \\ g_{1,2} = \epsilon_{1,2} \left[\sin(\Omega_2 \hat{t}) + \sin(\Omega_3 \hat{t})\right]^2 \left(\dfrac{1}{\left[1 - (y_1 + y_2)\right]^2} \pm \dfrac{1}{\left[1 - (y_1 - y_2)\right]^2} \right) \end{cases} \tag{18}$$

in which $\Omega_j = \tilde{\Omega}_j / \omega_{1s}$, and the excitation coefficients $(\mu, d_0, \epsilon_{1,2})$ are functions of nondimensional current $(\tilde{\mu})$ and voltage $(\tilde{\epsilon})$, mode shape coefficients, gap size and first-mode natural frequency.

2.4. State space formulation

For our investigation of the system's behaviour, we focus on the particular case of: $\Omega_1 = \dfrac{\Omega_2 + \Omega_3}{2}$. Hence, Eqs. (16) and (17) can be rewritten as the sixth-order autonomous system:

$$\begin{cases} \dot{y}_1 = w_1 \\ \dot{w}_1 = -y_1 - \alpha_{11} y_1^3 - \alpha_{12} y_1 y_2^2 - \eta_{11} y_1^2 w_1 - \eta_{12} y_1 y_2 w_2 + f_1(y_2, \psi_1) + g_1(y_1, y_2, \psi_1, \psi_2) - \zeta_1 w_1 \\ \dot{y}_2 = w_2 \\ \dot{w}_2 = -r^2 y_2 - \alpha_{21} y_1^2 y_2 - \alpha_{22} y_2^3 - \eta_{21} y_1 y_2 w_1 - \eta_{22} y_2^2 w_2 + g_2(y_1, y_2, \psi_1, \psi_2) - \zeta_2 w_2 \\ \dot{\psi}_1 = \Omega_1 \\ \dot{\psi}_2 = \Delta \end{cases} \tag{19}$$

wherein:

$$\begin{cases} f_1(y_2, \psi_1) = \mu \sin(\psi_1) \left[1 - \frac{1}{2} d_0^2 y_2^2\right] \\ \\ g_1(y_1, y_2, \psi_1, \psi_2) = 4\epsilon_1 \sin^2(\psi_1) \cos^2(\psi_2) \left(\dfrac{1}{\left[1 - (y_1 + y_2)\right]^2} + \dfrac{1}{\left[1 - (y_1 - y_2)\right]^2} \right) \\ \\ g_2(y_1, y_2, \psi_1, \psi_2) = 4\epsilon_2 \sin^2(\psi_1) \cos^2(\psi_2) \left(\dfrac{1}{\left[1 - (y_1 + y_2)\right]^2} - \dfrac{1}{\left[1 - (y_1 - y_3)\right]^2} \right) \end{cases}$$

and $\Delta = \Omega_3 - \Omega_1 = \Omega_1 - \Omega_2 = (\Omega_3 - \Omega_2)/2$. Is is thus clear that the state space in Eq. (19) is inherently governed by two frequencies, from which its response can be either periodic, quasi-periodic or chaotic as a function of the interplay between magnetic and electrodynamic excitations.

We note that the two-mode model proposed here is readily reduced to the single-mode approximation considered by Gottlieb et al. [26] by imposing $y_2 = 0$ in Eq. (19), which yields the fourth-order system:

$$\begin{cases} \dot{y}_1 = w_1 \\ \dot{w}_1 = -y_1 - \alpha_{11} y_1^3 - \eta_{11} y_1^2 w_1 + \mu \sin(\psi_1) + \dfrac{8\epsilon_1 \sin^2(\psi_1) \cos^2(\psi_2)}{\left[1 - y_1\right]^2} - \zeta_1 w_1 \\ \dot{\psi}_3 = \Omega_1 \\ \dot{\psi}_4 = \Delta \end{cases} \tag{20}$$

3. Estimation of parameters

To determine the values of the numerous non-dimensional parameters listed in the previous section, we take inspiration from the experiment by Scheible et al. [10]. The documented dimensions of the beam are: $(L, b, h) = (2, 0.2, 0.1)\mu m$, which lead to the non-dimensional curvature parameter $\tilde{\alpha} = 2400$. Estimating the capacitor gap to measure $\gamma = 40nm$, its non-dimensional version is $\tilde{\gamma} = 0.02$. We assume that the thin layer of conducting material used to coat the beam is negligible in terms of its mechanical properties, so that typical values for silicon ($\rho = 2300\ kg/m^3$, $E = 149\ GPa$) are chosen. Considering the maximum values for magnetic field and applied current, i.e. $B = 12T$ and $i = 19\mu A$, respectively, we deduce a limiting value of $\tilde{\mu} = 2.45 \times 10^{-4}$ for non-dimensional magneto-motive forcing amplitude. On the other hand, for unit voltage $V = 1V$ across the capacitors, $\tilde{\epsilon} = 2.38 \times 10^{-7}$.

3.1. Axial stress & nonlinear stiffness

The values of residual axial stress $\tilde{\beta}$ and effective torsional stiffness κ are not directly available from the data at hand. However, as a first-mode natural frequency of $f = 45.847\ MHz$ was reported for the experimental resonator [10], we may use this information to find the limiting values of both parameters. Firstly, in the ideal case of no residual stress ($\tilde{\beta} = 0$), the non-dimensional eigenfrequency is given by: $\hat{\omega}_1 = z_1^2$, where the elastic wave number z_1 is an implicit function of κ as per Appendix A. Fig. 3 shows the behaviour of $\hat{\omega}_1$ as a function of the latter, from which we deduce that the value $\kappa_0 = 0.38$ yields the correct eigenfrequency by considering the scaling frequency $\omega_s = 5.81\ Mrad/s$ for the given material and geometric properties [26]. It should be noted that this situation is highly idealised, as some residual stress due to fabrication defects is realistically bound to exist in the system.

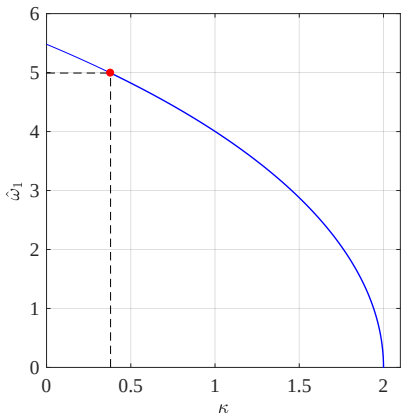


Figure 3: Evolution of non-dimensional first-mode natural frequency $\hat{\omega}_1$ with κ, for $\tilde{\beta} = 0$. The value $\kappa = 0.38$ corresponds the experimentally measured frequency.

Secondly, we consider the case where $\kappa \to \infty$, at which point the beam becomes doubly clamped at $s = \pm/2$ and no interaction is possible between the inner and outer sections. This has the effect of simplifying the compatibility conditions, Eq. (10), by discarding all equations involving fields B and $\bar{B}$ and including a zero-slope condition: $u_s^A(1/2, \tau) = u_s^{\bar{A}}(-1/2, \tau) = 0$. Having computed $z_1 = 2.341$ as the wave number for the doubly-clamped beam, the relation $\hat{\omega}_1^2 = z_1^4 + \tilde{\beta} I_1$ yields a value $\tilde{\beta}_0 = 2.175$ such that the dimensional eigenfrequency coincides with the measured value. In practice, we naturally must have $\tilde{\beta} < \tilde{\beta}_0$. It shall be noted that $\tilde{\beta}_0$ corresponds to a dimensional value of $N = 2.026\ \mu N$ in terms of axial force or $\sigma = 67.53\ MPa$ for axial stress, which is below the bulk modulus of silicon, $\sigma_M = 93\ MPa$. We remark that, through $\hat{\omega}_1^2$, the choice of $\tilde{\beta}$ also has an impact on the effective nonlinear curvature parameters, α_{ij}, as shown in Fig. 4.

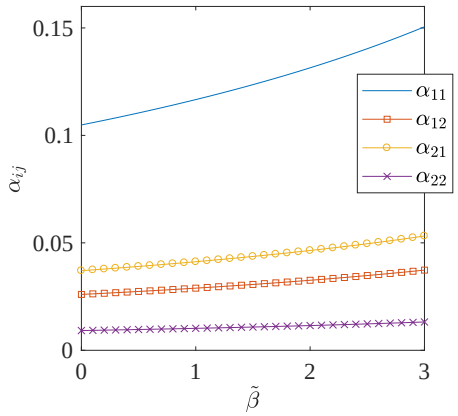


Figure 4: Dependence of nonlinear curvature parameters on $\tilde{\beta}$, given $\kappa = 0.42$.

Following the discussion in this section, we fix the values: $\kappa = 0.42$ and $\tilde{\beta} = 2.05$ for the remaining of this paper, which will be shown to yield a frequency ratio $r \approx 2$ corresponding to necessary conditions for a 2:1 internal resonance between the first asymmetric and symmetric modes. Additionally, this choice yields the following values for the curvature parameters, which we keep fixed throughout the following sections: $\alpha_{11} = 0.1332$, $\alpha_{12} = 0.0330$, $\alpha_{21} = 0.0471$, $\alpha_{22} = 0.0117$.

3.2. Linear damping

We estimate a reasonable range of damping based on [26]. Therein, the quality factor of the single-mode model, Q_1, was reported as: $10^2 < Q_1 < 10^4$. Given the relation:

$$Q_j = \frac{\hat{\omega}_j}{\zeta_j} \tag{21}$$

for the j-th mode and the fact that: $\hat{\omega}_{1s} = 1$, $\hat{\omega}_{1a} = r = 1.984$ for the current values of $(\kappa, \tilde{\beta})$, we readily find: $10^{-4} \leq \zeta_1 \leq 10^{-2}$ for the first mode damping and roughly twice these values for the second.
The impact of linear damping can be readily assessed by considering the near-resonant, single-mode response to purely magnetic excitation i.e. Eq. (20) with $\epsilon_1 = 0$. In the absence of nonlinear damping ($\eta_{11} = 0$) and fixing $\mu = 0.005$, the responses for $\zeta_1 = 5 \times 10^{-4}$ and $\zeta_1 = \times 10^{-3}$ are plotted together in Fig. 5.

As the system reduces to a periodically-forced Duffing oscillator in this case, a typical hardening behaviour is observed. In addition to increased maximum response amplitudes, the unstable branch delimited by two cyclic Saddle Node (SN) bifurcations extends farther in the low-damping scenario, leading to wider bi-stability regions. We note that, for $\zeta_1 = 5 \times 10^{-4}$, the resonant amplitude is close to $y_1 = 1$, the non-dimensional gap size.

3.3. Nonlinear damping

In order to estimate the nonlinear damping parameters, we consider the response of the single-mode model subjected to purely electrodynamic forcing, i.e. with $\mu = 0$. Setting $\Delta = 0$ in Eq. (20), it results that the system is -to first order of approximation- parametrically excited at the frequency $2\Omega_1$. Consequently, we compute the response in the

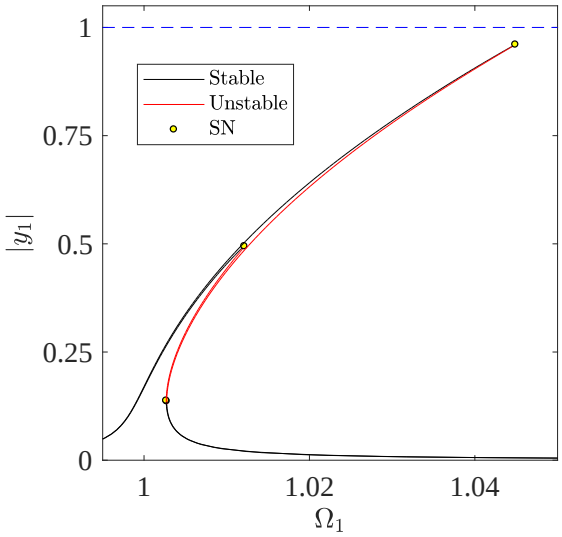


Figure 5: Frequency response of single-mode system to magnetic excitation with $\mu = 0.005$, for two values of linear damping: $\zeta_1 = 5 \times 10^{-4}$ and $\times 10^{-3}$. The ordinate in this plot corresponds to the maximum (absolute) value of y_1 over one oscillation period.

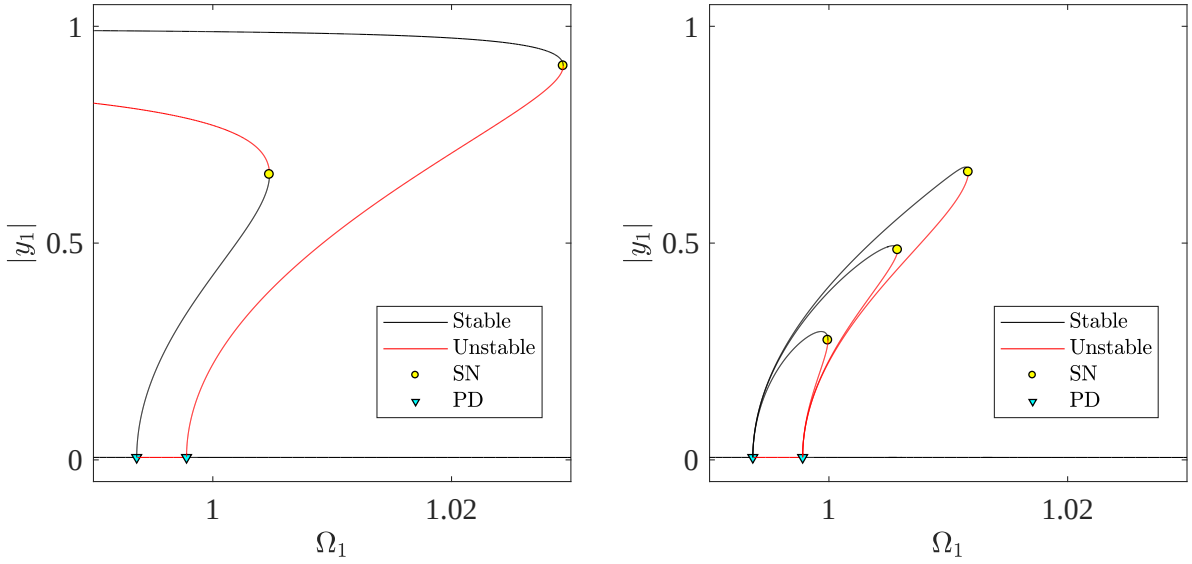


Figure 6: Principal parametric resonance of the single-mode system for $\epsilon_1 = 1.05 \times 10^{-3}$ and $\zeta_1 = 10^{-3}$, given zero (left) or nonzero (right) nonlinear damping, corresponding to: $\eta_{11} = \{0.1690, 0.0770, 0.0593\}$.

vicinity of the system's principal parametric resonance: $2\Omega_1 = 2$, or equivalently $\Omega_1 = 1$, while fixing $\epsilon_1 = 1.05 \times 10^{-3}$ and $\zeta_1 = \times 10^{-3}$. The result of this operation, with and without nonlinear damping, is shown in Fig. 6.

Parametric resonance manifests as a pair of super-critical/sub-critical Period Doubling (PD) bifurcations around the resonant frequency, which is shifted with respect to natural frequency. In the absence of nonlinear damping ($\eta_{11} = 0$), the bifurcating branches grow apart rather than meeting at a common SN bifurcation. As a consequence of the strong nonlinearity induced by the capacitors, this growth remains bounded and maximum amplitudes are restricted to the gap size. Nevertheless, such high-amplitude motions are non-physical in practical nanoresonators. To

resolve this issue, we include nonlinear damping by letting $\eta_{11} > 0$. As shown, this term effectively limits maximum amplitudes and causes the bifurcating branches to form a closed resonance peak. We choose the value of η_{11} in such a way that $|y_1|_{\max} < 0.5$, which is a relatively high but realistic limit. This condition is achieved for $\eta_{11} \leq 0.077$. The remaining coefficients can be easily deduced from this value since the ratios η_{ij}/η_{11} depend only on modal coefficients, as per Appendix C. Consequently, we find: $\eta_{12} = 0.0134$, $\eta_{21} = 0.0272$ and $\eta_{22} = 0.0047$.

4. Equilibrium Analysis

In the absence of external excitation and assuming no DC bias from the capacitors, Eq. (19) admits the unique fixed point: $\mathbf{z}_0 = [w_1, y_1, w_2, y_2] = [0, 0, 0, 0]$, which is stable for positive damping coefficients $\zeta_{1,2}$ and nonzero natural frequencies. We note that the latter are governed by two parameters: effective torsional stiffness κ and residual axial stress $\tilde{\beta}$, through the dispersion equations (Apendix A) and $\hat{\omega}_j^2 = z_j^4 + \tilde{\beta} I_j$. This defines a surface on the space $(\kappa, \tilde{\beta}, \hat{\omega}_j)$, as depicted in Fig. 7. For any fixed value of $\tilde{\beta}$, both natural frequencies decrease as κ increases, until $\hat{\omega}_1$ becomes zero and the beam buckles for a critical value $\kappa_{\text{lim}}(\tilde{\beta})$, approximately given by the affine relation: $\kappa_{\text{lim}}(\tilde{\beta}) = 2 - 0.51\tilde{\beta}$.

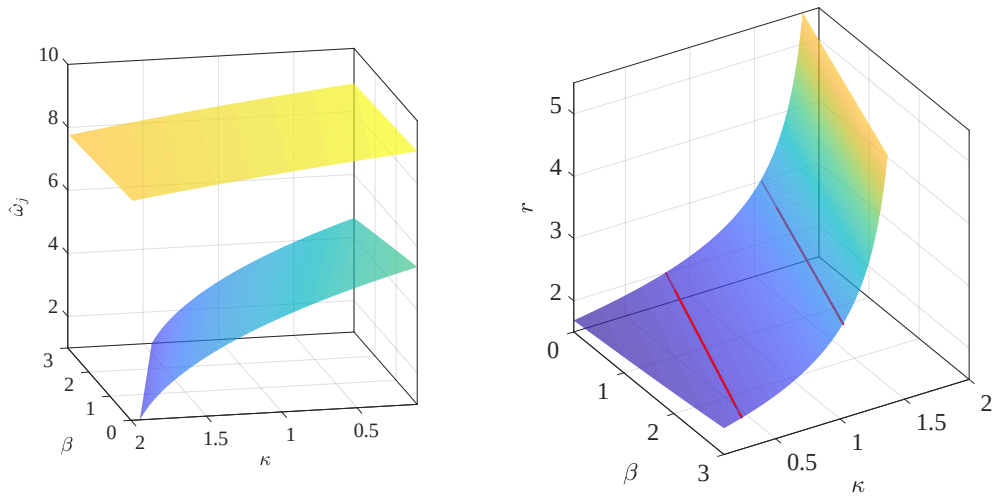


Figure 7: Natural frequencies as a function of $(\kappa, \tilde{\beta})$. Left: first (lower surface) and second (upper surface) modes. Right: frequency ratio, showing internal resonances 2:1 and 3:1 as red lines.

In the same figure we have represented the frequency ratio $r = \hat{\omega}_{1a}/\hat{\omega}_{1s}$. It immediately appears that r increases monotonically with both κ and $\tilde{\beta}$. Hence, the surface intersects the planes $r = m \in \mathbb{N}$ as it diverges asymptotically for $\kappa \to \kappa_{\text{lim}}$. This suggests the occurrence of internal resonances when the system is driven into the nonlinear regime. As an illustration of this phenomenon, Fig. 8 depicts a cross section of the above surfaces for $\tilde{\beta} = 0.38$. In that case, both 1:2 and 1:3 internal resonances can happen for small-to-moderate values of κ.

In the following sections, we characterise the system's dynamics by investigating its response to independently applied periodic inputs of either magneto-motive or capacitive nature, before considering the combined and quasi-periodic case.

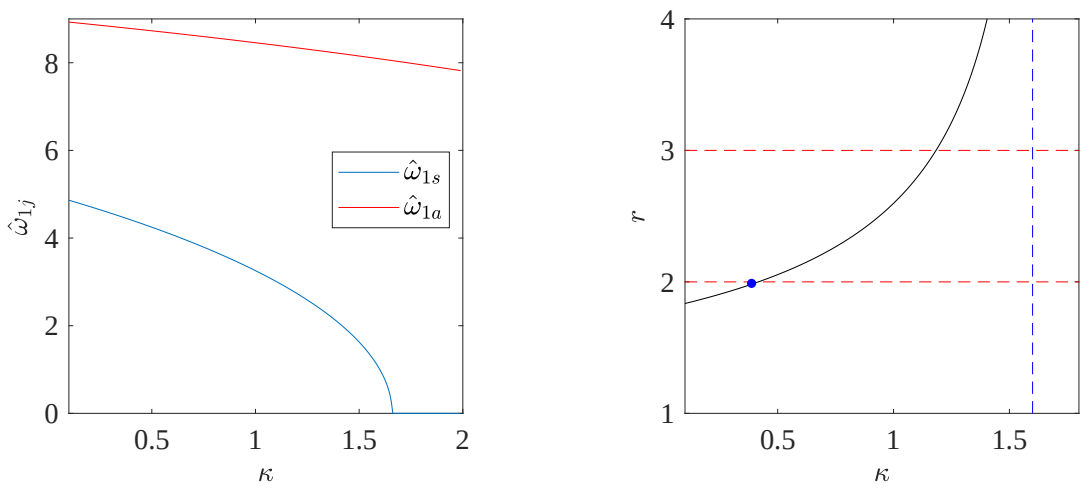


Figure 8: Impact of spring torsional stiffness on natural frequencies, for $\tilde{\beta} = 2.05$. Evolution of the ω_{1j} (left) and frequency ratio, r (right), showing potential internal resonances.

5. Response to Magnetic Excitation ($\epsilon_{1,2} = 0$)

We use pseudo arc-length continuation to construct frequency response curves. Steady-state periodic solutions are computed directly at each continuation step through a numerical implementation of the Harmonic Balance Method (HBM) [27], of which a brief overview is given in Appendix D. In this way, both stable and unstable branches are obtained along with any critical points found along the continuation process, such as cyclic Saddle Node (SN), Symmetry Breaking (SB) [28] or Period-Doubling [32] bifurcations. This approach is complemented by standard numerical integration (MATLAB's *ode45*) for validation and in-depth analysis of particularly interesting responses, when needed.

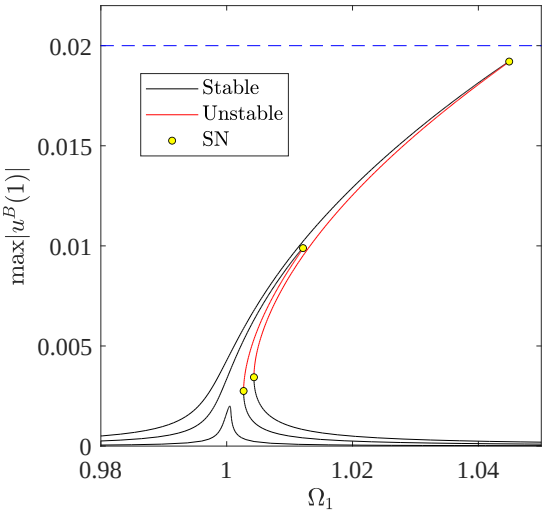


Figure 9: Frequency response of single-mode model for $Q = 10^3$ and increasing forcing amplitude: $\mu = 10^{-4}$, $\mu = 5 \times 10^{-4}$, $\mu = 10^{-3}$.

5.1. Single-mode dynamics

The system reduces to a periodically-forced nonlinear oscillator of the Duffing type, as shown in Fig. 9 for $Q = 10^3$ and varying forcing amplitudes: $\mu = 10^{-4}$, $\mu = 5 \times 10^{-4}$, $\mu = 10^{-3}$. Non-dimensional displacements at the beam tip are used as a measure, i.e.: $u^B(1) = \phi_{B1}(1)q_1(\tau) + \phi_{B2}(1)q_2(\tau) = \tilde{\gamma}\,(y_1(t) + y_2(t))$. The notation 'max' in this and subsequent figures shall be understood to represent the maximum value over one period of motion. For relatively strong forcing, the usual behaviour of systems with hardening nonlinearities is observed, with a region of bi-stability bounded by SN bifurcations connected by a branch of unstable solutions, giving rise to amplitude jumps upon ascending or descending frequency sweeps. Conversely, a virtually linear response is obtained in the case $\mu = 10^{-4}$.

5.2. Two-mode dynamics

Inspection of Eq. (19) readily reveals that solely the first mode is directly driven, whereas the second may only activate through nonlinear coupling. As a result, second-mode motions tend to vanish over an initial transient, leading to steady state responses with only first-mode motions over most of the frequency-response curve. This is verified through numerical simulation, which yields a curve indistinguishable from its single-mode counterpart.

However, the system exhibits symmetry-breaking bifurcations when subjected to high-amplitude forcing, such as $\mu = 0.2$. While this would result in maximal response amplitudes greater than the gap size γ, allowable amplitudes can be obtained by increasing the strength of nonlinear damping. For instance, by choosing $\eta_{11} = 1.54$ and adjusting the remaining nonlinear damping parameters accordingly yields the response of Fig. 10.

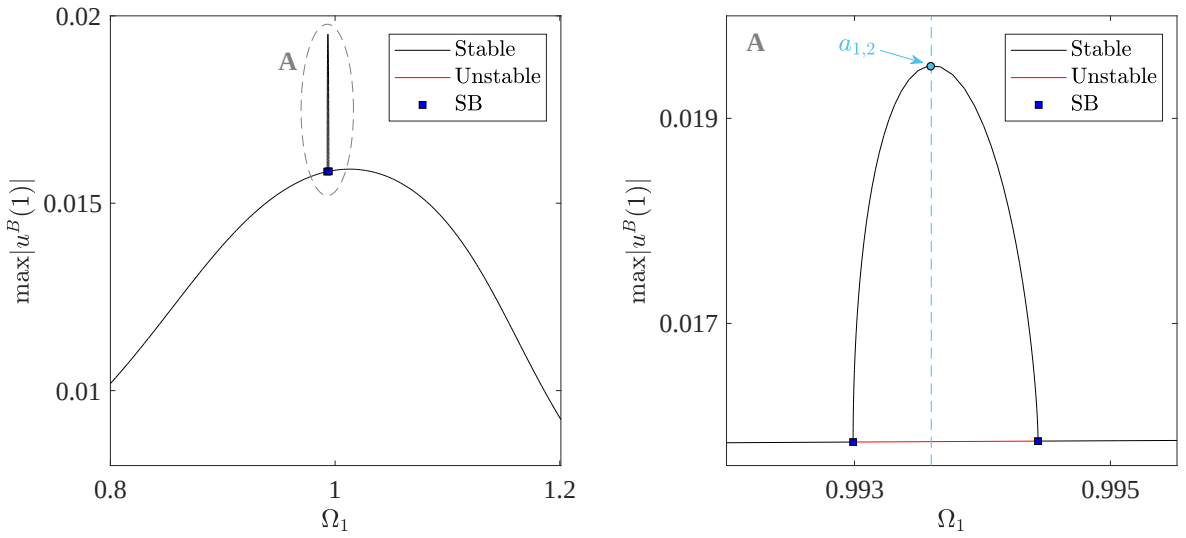


Figure 10: Frequency response of the two-mode model to magnetic excitation, for $\mu = 0.1$ and $\eta_{11} = 1.54$. Global view (left) and close up-view of region **A** containing SB bifurcations (right).

We note that the equations of motion, Eqs. (16) and (17), are invariant with respect to the change of variables: $(y_1, y_2) \rightarrow (y_1, -y_2)$. Thus, for any solution with non-zero contribution of the second mode, a conjugate cycle with identical amplitude but inverted orientation in phase space is also a solution. This behaviour is verified by numerically integrating the equations of motion at $\Omega_1 = 0.9936$ and using different initial conditions, which yields the two coexisting cycles shown in Fig. 11

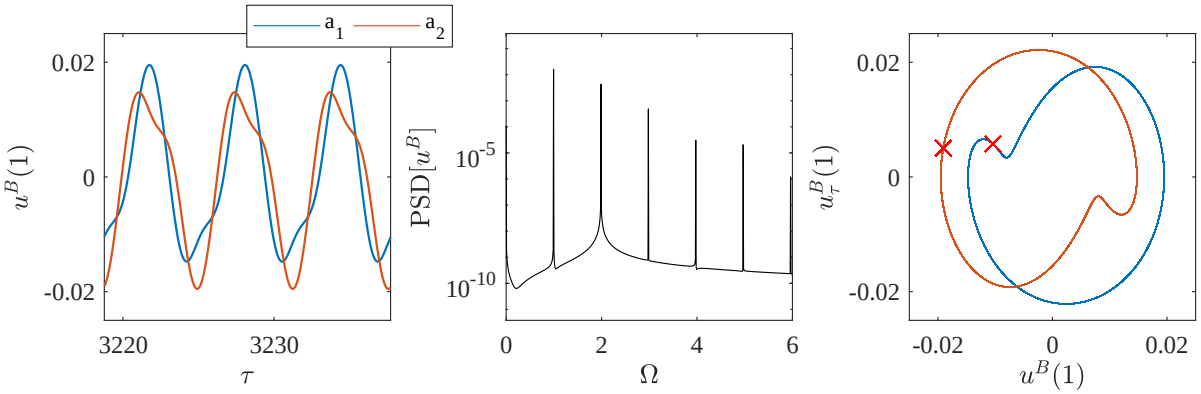


Figure 11: Conjugate two-mode solutions generated by symmetry breaking at $\Omega_1 = 0.9936$: Time histories (left), spectral density (center) and phase space representation (right).

6. Response to Electrodynamic Excitation ($\mu = 0$)

Inducing a periodic input signal by setting $\Delta \to 0$, which implies $\Omega_2 = \Omega_3 = \Omega_1$, modal electrodynamic excitation then takes the form:

$$g_{1,2}(\hat{t}, y_1, y_2) = 2\epsilon_{1,2}\left[1 - \cos(2\Omega_1\hat{t})\right]\left[\frac{1}{\left[1-(y_1+y_2)\right]^2} \pm \frac{1}{\left[1-(y_1-y_2)\right]^2}\right] \tag{22}$$

where, as before, the plus and minus signs corresponds to the first- and second-mode equations, respectively. Throughout this section, the quality factors are fixed at: $Q_1 = 10^3$ and $Q_2 = 3151$.

Before investigating the system's dynamics numerically, we consider the qualitative effect of a force with the form of Eq. (22). Introducing this expression into the equation of motion for the single-mode model yields:

$$\ddot{y}_1 + \zeta_1\dot{y}_1 + y_1 + \alpha_{11}y_1^3 - \frac{4\epsilon_1}{\left[1-y_1\right]^2} = -4\epsilon_1\frac{\cos(2\Omega_1\hat{t})}{\left[1-y_1\right]^2} \tag{23}$$

Assuming y_1 to be small, a third-order Taylor expansion can be used to approximate the rational function $(1-y_1)^2 = 1 + 2y_1 + 3y_1^2 + 4y_1^3 + O(|y_1|^4)$, which consequently simplifies Eq. (23) to:

$$\ddot{y}_1 + \zeta_1\dot{y}_1 + (1-4\epsilon_1)y_1 - 12\epsilon_1 y_1^2 + (\alpha_{11} - 16\epsilon_1)y_1^3 = 4\epsilon_1\left[1 - \cos(2\Omega_1\hat{t})(1 + 2y_1 + 3y_1^2 + 4y_1^3)\right] + O(|y_1|^4) \tag{24}$$

Inspection of the left-hand side terms of this equation reveals several distinct features, as follows:

- Detuning of the resonance frequency for $\epsilon_1 \neq 0$, as seen from the linear term. In particular, as ϵ_1 is strictly positive, the frequency is decreased (softening effect).
- Presence of a quadratic nonlinearity. This implies superharmonic resonances for $2\Omega_1 \ \omega_m/k$, where $k > 0$ is an even integer and $\omega_m^2 = 1 - 2\epsilon_1$. Likewise, this term can be responsible for unbounded oscillations [30], which physically corresponds to dynamic pull-in [31] in the present context as the system response exceeds the allowable capacitor gap, i.e. $|u^B(\pm 1)| = \tilde{\gamma}$ or $y_1 + y_2 = 1$.

- Modified cubic stiffness. As $\epsilon_1 \to 0$, the system behaves as a hardening Duffing oscillator. Conversely, as the voltage amplitude grows, the cubic stiffness $(\alpha_{11} - 16\epsilon_1)$ decreases and may even change sign, leading to a softening nonlinearity. This is significant, in particular, for second-mode dynamics, since α_{22} is an order of magnitude lower than α_{11} and thus more sensitive to changes in ϵ_1.

In addition, the right-hand side terms can be split into three distinct components:

- Shift in the mean oscillation amplitude, from 0 to $4\epsilon_1$.
- Periodic excitation $4\epsilon(2\Omega_1\hat{t})$, leading to a principal resonance near $\Omega_1 = \omega_m/2$.
- Parametric excitation $4\epsilon(2\Omega_1\hat{t})(2y_1 + ...)$, implying parametric resonance near $\Omega_1 = \omega_m$ with double-period responses.

The non-dimensional amplitude parameters $\epsilon_{1,2}$ effectively pilot the dynamics of the system along with the excitation frequency $2\Omega_1$. Following the definitions introduced in Appendix C, these amplitudes have the same dependence to input voltage and thus do not vary independently but are proportional to each other. Thus, it is sufficient to consider the variation of ϵ_1 and adjust ϵ_1 accordingly.

6.1. Single-mode dynamics

We focus on the main parametric resonance of the first mode, near $2\Omega_1 \approx 2$. Following Sect. 6, we choose $\eta_{11} = 0.169$ and $\epsilon_1 = 1.05 \times 10^{-3}$, which produces the frequency response plot shown in Fig. 12. Near twice the detuned natural frequency, a resonance peak consisting of a stable and an unstable branch emerges from a pair of PD bifurcations, labelled *PD*1 and *PD*2 in the figure. Both branches meet at a cyclic SN bifurcation, labelled *LP*1. The same figure displays the result of tracking these bifurcations in the parameter plane (Ω_1, ϵ_1). Therein, we observe that all three of these bifurcations are born from the same point at the critical value ϵ_1^{GPD}, which thus corresponds to a so-called *generalised period doubling* bifurcation [32].

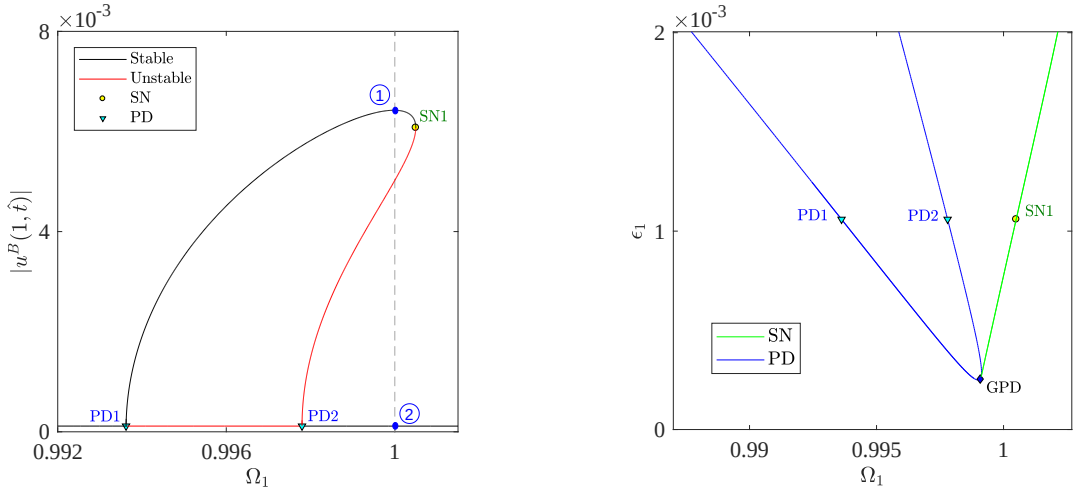


Figure 12: Main parametric resonance of the first mode: frequency response for $\eta_{11} = 0.169$ and $\epsilon_1 = 1.05 \times 10^{-3}$ (left), bifurcation tracking in (Ω_1, ϵ_1) plane.

The qualitative differences between the coexisting cycles on the parametric and main response branches can be clearly seen in Fig. 15, where we have used numerical integration at $\Omega_1 = 1$.

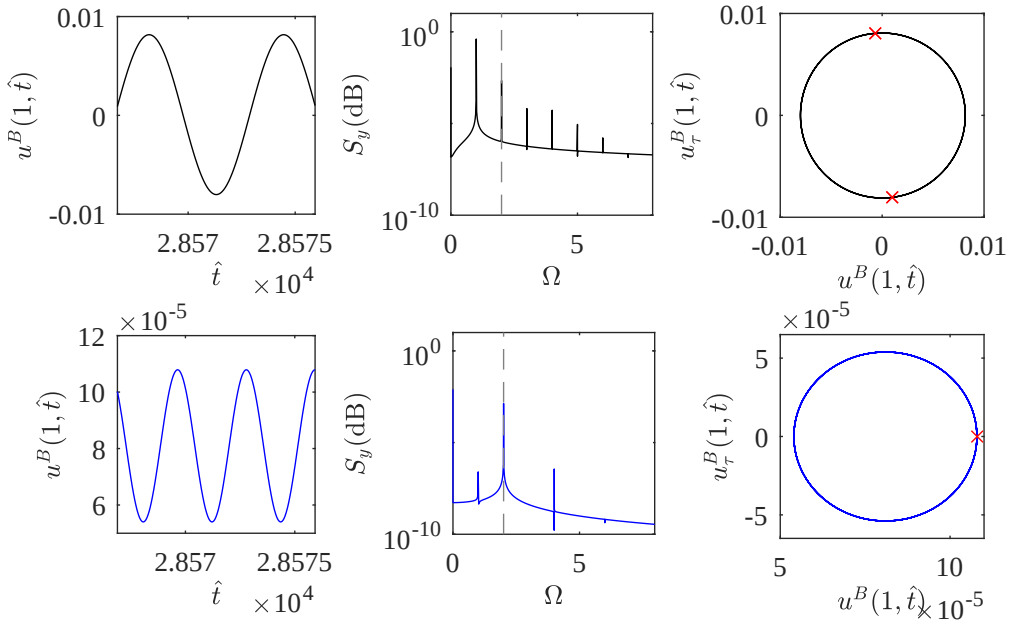


Figure 13: Time histories (left), power spectra (centre) and phase-space plots (right) for the two coexisting solutions at $\Omega_1 = 1$, $\epsilon_1 = 1.7 \times 10^{-4}$, labelled ① and ② in Fig. 12. Top: high-amplitude, double-period cycle on resonant branch; bottom: low-amplitude cycle on main response curve. Dashed lines on spectra show the forcing frequency, markers on phase plane correspond to fixed points of the associated Poincare map.

6.2. Two-mode dynamics

When a second mode is included in the model, there is an additional parametric resonance of the aforementioned near $2\Omega_1 = 2r \approx 4$, or $\Omega_1 \approx 2$, with two new period doubling bifurcations manifesting on the response curve. Inducing this phenomenon requires excitation amplitude to be increased; in the left-most plot of Fig. 14 we have used $\epsilon_1 = 2.12 \times 10^{-3}$.

In contrast to the previous parametric resonance, cycles on the bifurcated branches are largely dominated by the second mode, as can be appreciated from the numerical integration results of Fig. 15. Thus, nonlinear damping effects are piloted by η_{22} rather than η_{11}. However, from Sec. 3.3, we know that the former is an order of magnitude smaller than the latter. This is why, for the same nonlinear damping coefficients as before, the bifurcating branches do not intersect in Fig. 14. Increasing η_{11} from 0.0770 to 1.540, which in turn takes η_{22} from 0.0047 to 0.0946, provides sufficient damping so that a closed resonance peak with maximum amplitude lower than half the gap size, $\gamma = 0.02$, is obtained. This is shown in the same figure, next to responses for intermediate values of η_{11}.

For the most damped case, we have tracked bifurcations with respect to ϵ_1, which yields the bifurcation diagram shown at the right in Fig. 14. When compared to the corresponding plot for the single-mode case, i.e. Fig. 12, we remark several dissimilar features. SN bifurcations are not born in parallel to the pair of PD's, but rather at a higher forcing amplitude value ϵ_1^{SNPD}. Thus, response curves below this value consist of a single stable branch, as per Fig. 16. Moreover, there is a local maximum, or cusp (marked CSP), in the SN curve which indicates the maximum value for which the stable and unstable branches join at a saddle-node bifurcation.

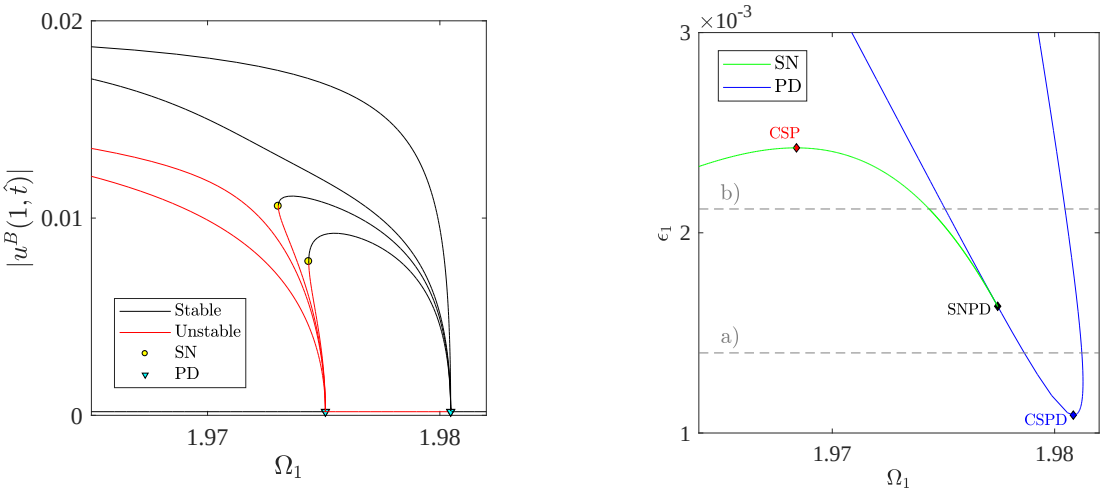


Figure 14: Main parametric resonance of the second mode: frequency response for $\epsilon_1 = 2.12 \times 10^{-3}$ and four different values of the nonlinear damping parameter η_{11}, namely: 0.077, 1.155, 1.309 and 1.54, of which the two highest yield closed resonance peaks (left); bifurcation tracking for $\eta_{11} = 1.54$.

7. Response to Combined Excitation

In this section, we consider two different but close excitation frequencies: $\Omega_2 = 0.999$, $\Omega_3 = 1.001$, which match the experimental values $f_2 = 45.802\ MHz$, $f_3 = 45.892\ MHz$ [10]. As this is equivalent to $\Omega_2 = \Omega_1 - \Delta$ and $\Omega_3 = \Omega_1 + \Delta$, the frequency Ω_1 is therefore equal to 1 and the forcing is quasi-periodic. Likewise, the non-dimensional amplitude of electrodynamic excitation is fixed at: $\tilde{\epsilon} = 2 \times 10^{-6}$, which yields the modal amplitudes: $\epsilon_1 = 0.0011$ and $\epsilon_2 = 0.0423$.

7.1. Single-mode dynamics

The response is studied by numerically integrating Eq. (20) for different combinations of μ, ϵ_1 and Q_1, and three types of results are presented: time histories, power spectral densities (PSD), and Poincaré maps. For the latter, the steady-state displacements and velocities are sampled at intervals equal to the period of magnetic forcing:

$$u_{Bn}(1) = u^B(1, \tau_n) \quad , \quad \tau_n = \tau_0 + \frac{n}{2\pi}, \quad n \in \mathbb{N}$$

where τ_0 is an arbitrary instant belonging to the steady regime. A total of 2^{10} samples was used to generate the results shown in Fig. 19, where $Q_1 = 1000$.

A torus doubling bifurcation is immediately evident as a slight magnetomotive force is considered, as seen in the transition from the first to the second rows. While the former of these has a spectrum consisting of a series of peaks centered around integer multiples of $2\Omega_1$, the latter also includes frequency components around odd multiples of Ω_1. Furthermore, even though the magnetomotive signal with frequency Ω_1 is relatively weak, considerable mechanical mixing is achieved around said frequency under the present configuration. This effect is enhanced with increasing electromotive amplitude ϵ_1, as seen in the spectrum of the last row.

By taking once more $\epsilon_1 = 0.0011$ and $\mu = 0.0001$, we now consider the effect of increasing the quality factor to $Q_1 = 10000$. This leads to erratic, chaotic-like oscillations over extended periods, as seen in Fig. 18. This chaotic

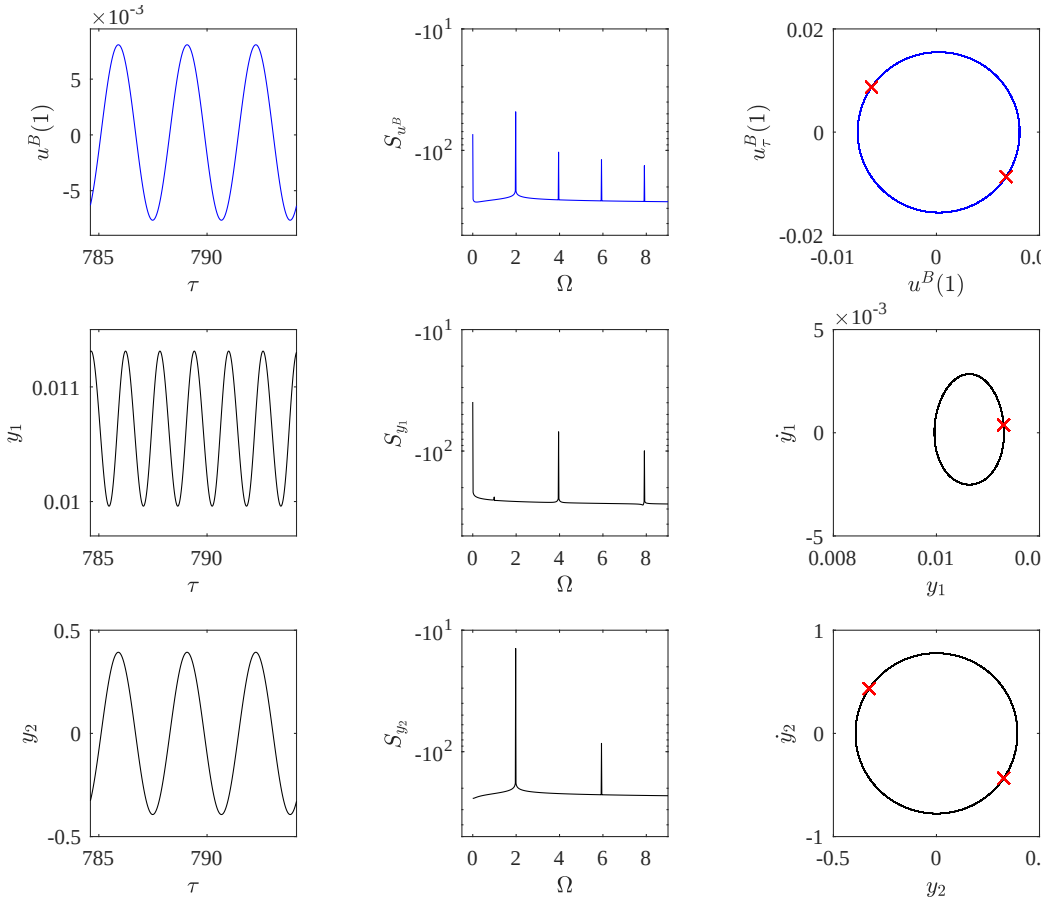


Figure 15: Time histories (left), power spectra (centre) and phase-space plots (right) for the periodic solution at $\Omega_1 = 1.976$, $\epsilon_1 = 2.12 \times 10^{-3}$, $\eta_{11} = 1.540$. Beam-tip displacement $u^B(1, \hat{t})$ (top) and modal components y_1 (middle) and y_2 (bottom).

response is transient in nature, as integrating over a sufficiently long time span leads to stabilization of the response into a quasi-periodic regime.

The presence of transient chaos is subjected to having high quality factors and, everything else being equal, its duration is proportional to the value of ϵ_1, as illustrated by Fig. 19.

7.2. Two-mode dynamics

For excitation in the range discussed previously, the addition of a second mode has no perceptible effect on the response to combined excitation. By adjusting the input frequencies so that $\delta = 0.001$ and $\Omega_1 = r$, we are able to obtain two-mode responses in which, analogously to Sect. 6.2, the system has undergone torus doubling. An example of this is shown in Fig. 20 for $Q_1 = 1000$, $\mu = 0.0001$ and $\epsilon_1 = 0.0022$. On the other hand, no occurrence of chaotic transients was observed.

8. Discussion

A number of physical mechanisms, contribute to the rich dynamical behaviour of the nanoresonator considered in the present study. The results gathered cast light on the relative importance of each of these as they pertain to the

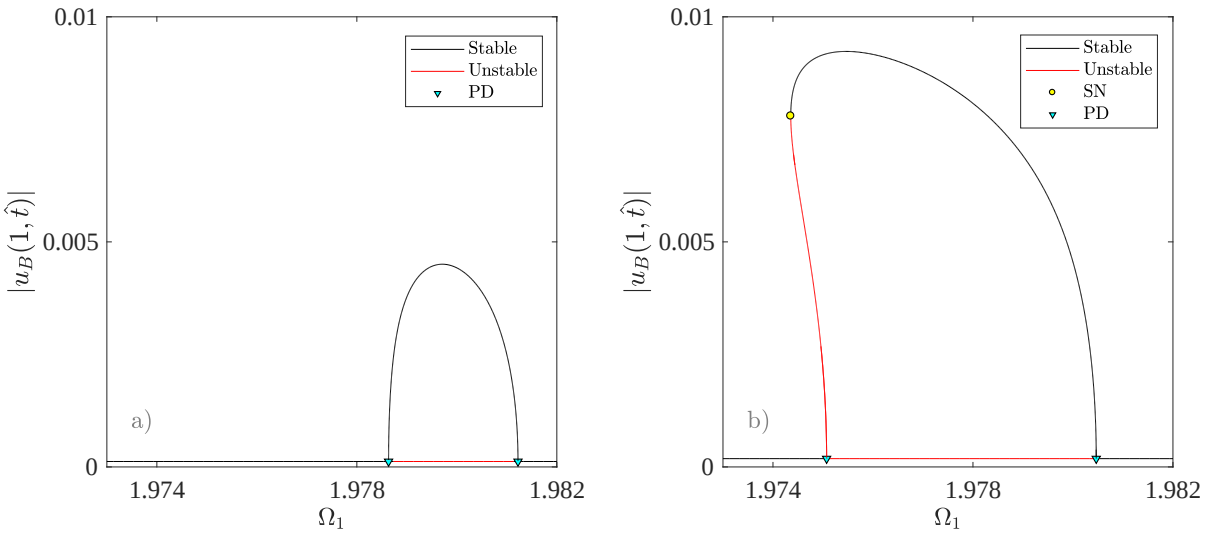


Figure 16: Main parametric resonance of second mode for $\epsilon_1 = 1.4\times10^{-3}$ (a) and 2.1×10^{-3} (b), respectively lower and higher than the bifurcation threshold ϵ_1^{SNPD}.

mathematical model employed, and suggest directions for further work.

Nonlinear damping, in particular, was numerically shown to have a major effect on the resonant responses, whether the single or multiple-mode models were considered. While our model correctly captures the qualitative behaviour observed in previous experiments, a fine fitting of dissipative forces from empirical data is deemed necessary for design purposes.

The multi-mode nature of the present model is another aspect that the authors consider worthy of further investigation. As shown from the results of Secs. 5 through 7, the consideration of a second mode broadens the spectrum of observable behaviours. While here the principal identified mechanism of modal interaction was symmetry breaking, internal resonances may become important when even higher modes are taken into account, which is especially relevant if higher-frequency excitations are to be targeted.

Finally, the bifurcation analysis of the quasi-periodic regimes is an interesting perspective to which the HBM-based algorithms used here lend themselves quite naturally, if extended to work on a two-frequency Fourier basis. In this manner, an efficient optimization of a given nanoresonator can be achieved.

9. Concluding remarks

We have derived and analysed a quasi-continuum initial-boundary-value problem for a three-field nanoresonator that consistently incorporates geometric nonlinearity of the elastic centre-field with the nonlinear contributions of magneto-motive and electrodynamic excitation. Symmetry and modal projection were used to reduce the spatio-temporal field equations to a low-order, equivalent nonlinear dynamical system under quasi-periodic excitation. The governing parameters were estimated to match the resonance conditions of the experimental setup by Scheible et al. [10]. Numerical analysis reveals interactions in the vicinity of 2:1 internal resonance between the first symmetric and anti-symmetric modes, leading to symmetry breaking, in the case of pure, high-amplitude magneto-motive forcing.

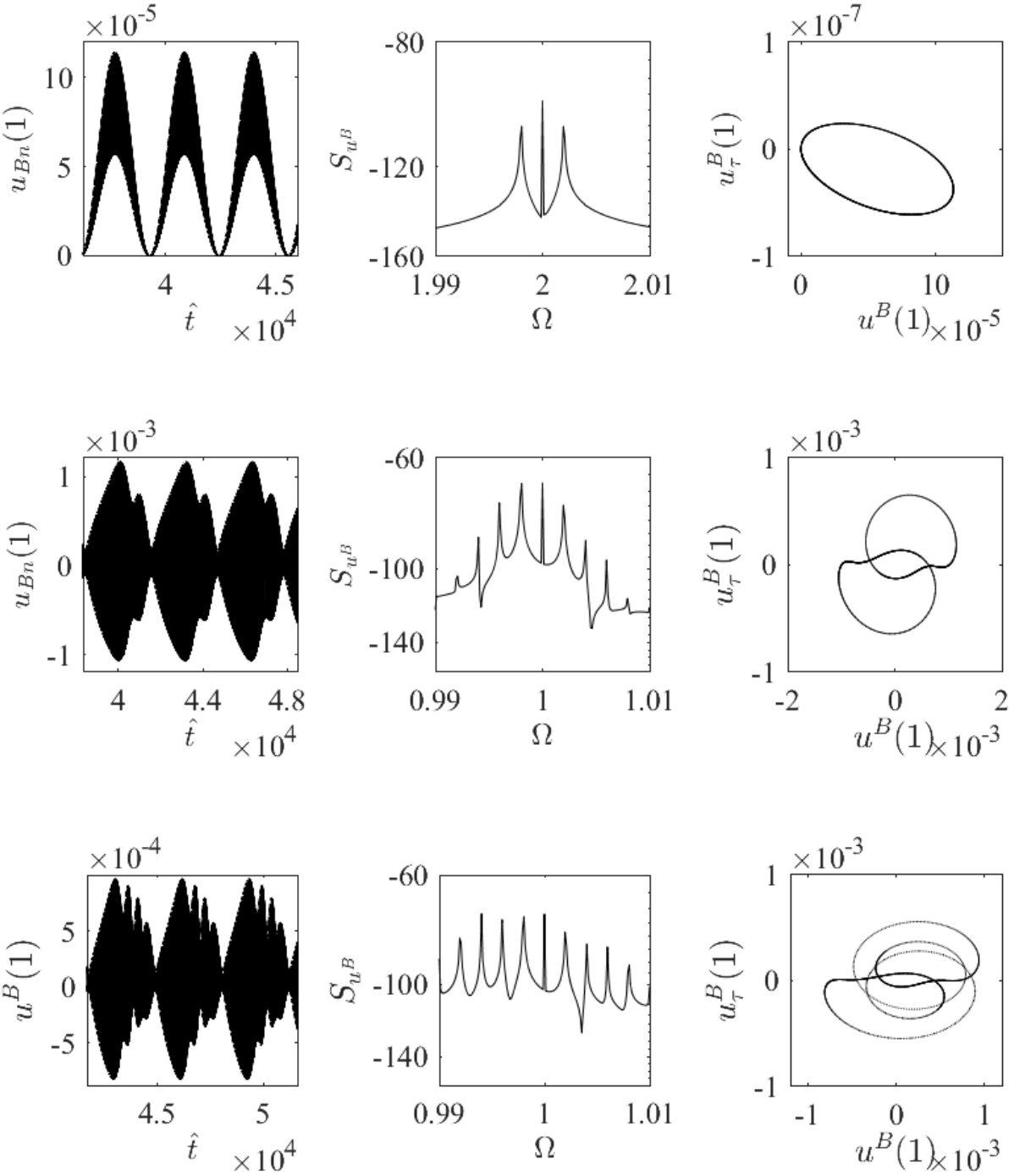


Figure 17: Response to combined magnetic and electrodynamic excitation at right-most beam tip, $s = 1$, for different parameter values: a) $\mu = 0$, $\epsilon_1 = 0.0011$, b) $\mu = 0.0001$, $\epsilon_1 = 0.0011$, c) $\mu = 0.0001$, $\epsilon_1 = 0.0022$. Left column: time history, centre column: spectral density, right column: Poincaré map.

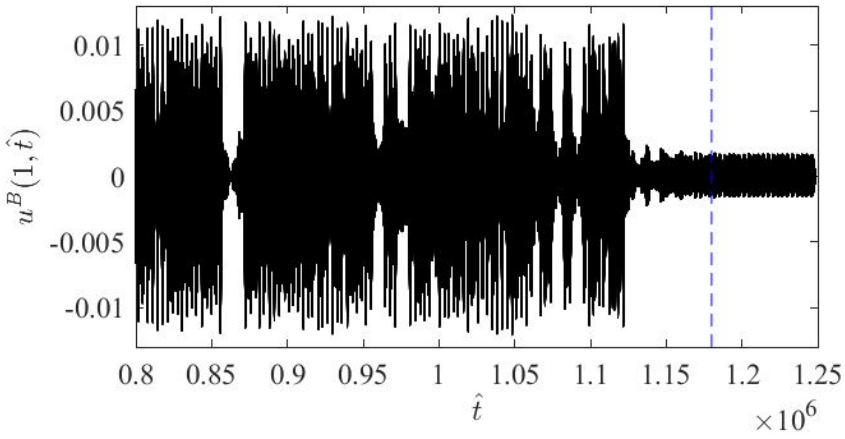


Figure 18: Transition from transient chaos to stable quasi-periodic orbit, $Q_1 = 10^4$.

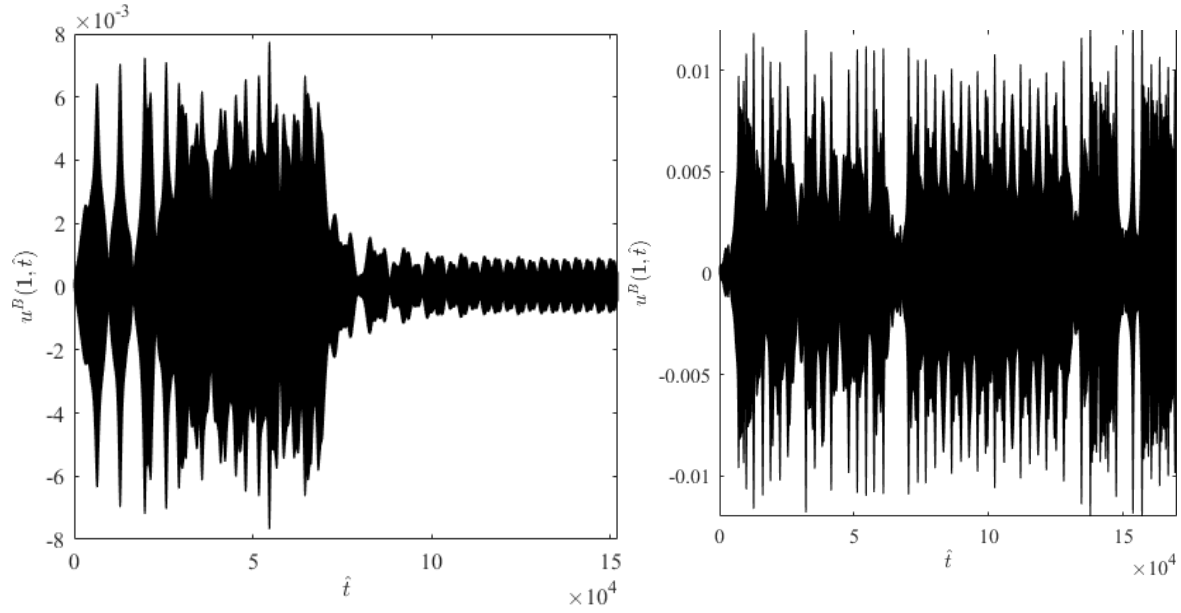


Figure 19: Time history of beam-tip response to combined excitation for $\mu = 0.0001$, $Q_1 = 10000$ and two values of ϵ_1: 5.3×10^{-4} (left) and 0.0011 (right).

With the addition of electrodynamic excitation, quasi-periodic responses with increasing complexity were observed, as evidenced by increasingly rich power spectra suggesting potential transition to chaos. Future work will focus on construction of the complete bifurcation structure and the investigation of sensitivity of the reduced dynamical system.

Acknowledgements

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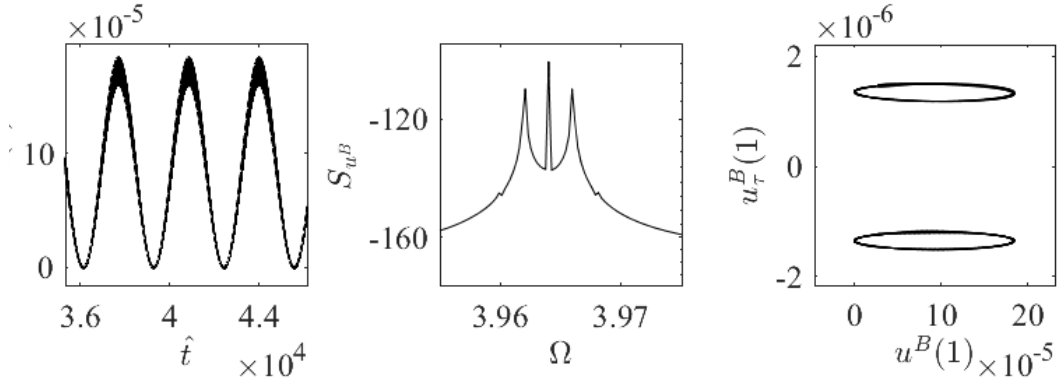


Figure 20: Response of two-mode system to combined excitation near second parametric resonance, given: $Q_1 = 1000$, $\mu = 0.0001$ and $\epsilon_1 = 0.0022$.

Appendix A. Eigenmode computation

Each displacement field is approximated by its first symmetric (index '1') and anti-symmetric (index '2') bending modes:

$$\begin{cases} u^A(s,\tau) & = \phi_1^A(s)q_1(\tau) + \phi_2^A(s)q_2(\tau) \quad , \quad s \in [0, 1/2] \\ u^B(s,\tau) & = \phi_1^B(s)q_1(\tau) + \phi_2^B(s)q_2(\tau) \quad , \quad s \in [1/2, 1] \\ u^{\bar{A}}(s,\tau) & = \phi_1^{\bar{A}}(s)q_1(\tau) + \phi_2^{\bar{A}}(s)q_2(\tau) \quad , \quad s \in [-1/2, 0] \\ u^{\bar{B}}(s,\tau) & = \phi_1^{\bar{B}}(s)q_1(\tau) + \phi_2^{\bar{B}}(s)q_2(\tau) \quad , \quad s \in [-1, -1/2] \end{cases} \tag{A.1}$$

The mode shape functions ϕ_j^X are found by introducing the modal ansatz $u^X(s,\tau) = e^{i\omega_j\tau}\phi_j^X$ into the underlying undamped, unforced linear problem with $\tilde{\beta} = 0$, which is of the form:

$$u^X_{ssss} + u^X_{\tau\tau} = 0 \tag{A.2}$$

for $X = \left\{\bar{B}, \bar{A}, A, B\right\}$. It immediately follows that the right half-span functions are given by:

$$\phi_i^A(s) = A_{1i}\cosh(z_i s) + A_{2i}\cos(z_i s) + A_{3i}\sinh(z_i s) + A_{4i}\sin(z_i s) \quad s \in [0, 1/2]$$
$$\phi_i^B(s) = B_{1i}\cosh(z_i s) + B_{2i}\cos(z_i s) + B_{3i}\sinh(z_i s) + B_{4i}\sin(z_i s) \quad s \in [1/2, 1]$$

where $z_i^4 = \omega_i^2$ is the elastic wave number corresponding to the i-th mode. Note that the ω_i should not be confused with the natural frequencies $\hat{\omega}_i$, as the two only coincide in the absence of axial stress. While analogous expressions can be written for the mode shape functions over the left half-span, it is more convenient to exploit the symmetry of any given mode and express everything as a function of the above set of eight coefficients $\{A_{1i}, ..., A_{4i}, B_{1i}, ..., B_{4i}\}$. These are found by replacing $u^X(s,\tau) = e^{i\hat{\omega}_j\tau}\phi_j^X$ into Eqs. (9), (10) and either (12) or (**??**) if symmetric or anti-symmetric modes are sought, respectively. We note that, in the former case, the symmetry condition at $s = 0$ imposes $A_{3i} = A_{4i} = 0$ for all i, whereas in the latter case $A_{1i} = A_{2i} = 0$. Hence, by letting: $\boldsymbol{\phi}_{si} = [A_{1i}, A_{2i}, B_{1i}, B_{2i}, B_{3i}, B_{4i}]^t$ and $\boldsymbol{\phi}_{ai} = [A_{3i}, A_{4i}, B_{1i}, B_{2i}, B_{3i}, B_{4i}]^t$ contain the coefficients for each mode type, we are led to the following two linear problems to be solved for the six mode shape coefficients and elastic wave number in each case:

$M_\kappa^S(z)\boldsymbol{\phi}_1 = \mathbf{0}$, where:

$$M_\kappa^S(z)\begin{bmatrix} 0 & 0 & CH_z & -C_z & SH_z & -S_z \\ 0 & 0 & SH_z & S_z & CH_z & -C_z \\ CH_{\frac{z}{2}} & C_{\frac{z}{2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & CH_{\frac{z}{2}} & C_{\frac{z}{2}} & SH_{\frac{z}{2}} & S_{\frac{z}{2}} \\ SH_{\frac{z}{2}} & -S_{\frac{z}{2}} & -SH_{\frac{z}{2}} & S_{\frac{z}{2}} & -CH_{\frac{z}{2}} & -C_{\frac{z}{2}} \\ CH_{\frac{z}{2}} - \frac{\kappa}{2}SH_{\frac{z}{2}} & -C_{\frac{z}{2}} + \frac{\kappa}{2}S_{\frac{z}{2}} & -CH_{\frac{z}{2}} & C_{\frac{z}{2}} & -SH_{\frac{z}{2}} & S_{\frac{z}{2}} \end{bmatrix} \tag{A.3}$$

and:

$M_\kappa^A(z)\boldsymbol{\phi}_2 = \mathbf{0}$, where:

$$M_\kappa^A(z) = \begin{bmatrix} 0 & 0 & CH_z & -C_z & SH_z & -S_z \\ 0 & 0 & SH_z & S_z & CH_z & -C_z \\ SH_{\frac{z}{2}} & S_{\frac{z}{2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & CH_{\frac{z}{2}} & C_{\frac{z}{2}} & SH_{\frac{z}{2}} & S_{\frac{z}{2}} \\ CH_{\frac{z}{2}} & C_{\frac{z}{2}} & -SH_{\frac{z}{2}} & S_{\frac{z}{2}} & -CH_{\frac{z}{2}} & -C_{\frac{z}{2}} \\ SH_{\frac{z}{2}} - \frac{\kappa}{2}CH_{\frac{z}{2}} & -S_{\frac{z}{2}} - \frac{\kappa}{2}C_{\frac{z}{2}} & -CH_{\frac{z}{2}} & C_{\frac{z}{2}} & -SH_{\frac{z}{2}} & S_{\frac{z}{2}} \end{bmatrix} \tag{A.4}$$

in which the short-hand notations $CH_z = \cosh(z)$, $C_z = \cos(z)$, and so forth have been used convenience. A nontrivial solution to the above systems can only exist if the matrices have null determinants, which is equivalent to finding zeros of the dispersion relations:

$$\begin{aligned} \det\left(M_\kappa^S\right)(z) =& 2z\cos(z) + \cosh(z/2)\left[8z\cos(z/2) - 4\kappa\sin(z/2)\right] - \kappa\sin(z) \\ &+ \cosh(z)(2z + 4z\cos(z) - \kappa\sin(z)) - \\ &4\kappa\cos(z/2)\sinh(z/2) - \kappa\sinh(z)(1 + \cos(z)) = 0 \end{aligned} \tag{A.5}$$

and:

$$\begin{aligned} \det\left(M_\kappa^A\right)(z) =& 2z\cos(z) - 4\kappa\cosh(z/2)\sin(z/2) + \kappa\sinh(z)(1 + \cos(z)) \\ &- \kappa\sin(z) - \cosh(z)(2z + \kappa\sin(z)) + 4\kappa\cos(z/2)\sinh(z/2) + \\ &8z\sin(z/2)\sinh(z/2) + 4z\sin(z)\sinh(z) = 0 \end{aligned} \tag{A.6}$$

Infinite solutions exist for the transcendental Eqs. (A.5) and (A.6), which occur in the order: $z_{si} < z_{ai} < z_{s(i+1)}$ as can be appreciated in Table A.1 for the first three pairs of modes.

Having found the wave number for a given mode, z_i, Eqs. (A.3) and (A.4) evaluated at $z = z_i$ and augmented with

	S_1	A_1	S_2	A_2	S_3	A_3
z	2.2343	2.9822	4.3343	6.805	8.3419	8.8702
ω	4.9921	8.8935	18.7862	46.3080	69.5873	78.6804

Table A.1: Non-dimensional roots to characteristic equations and corresponding eigenfrequencies for $\tilde{\beta} = 0$.

the normalization condition:

$$\int_0^{\frac{1}{2}} \left(\phi_{Aj}\right)^2 ds + \int_{\frac{1}{2}}^{1} \left(\phi_{Bj}\right)^2 ds + \int_{-\frac{1}{2}}^{0} \left(\phi_{\bar{A}j}\right)^2 ds + \int_{-1}^{-\frac{1}{2}} \left(\phi_{\bar{A}j}\right)^2 ds = 1 \tag{A.7}$$

uniquely determine the shape coefficients. For $\kappa = 0.38$, the result of this operation yields the shapes shown in Fig. A.21 for the first three pairs of modes.

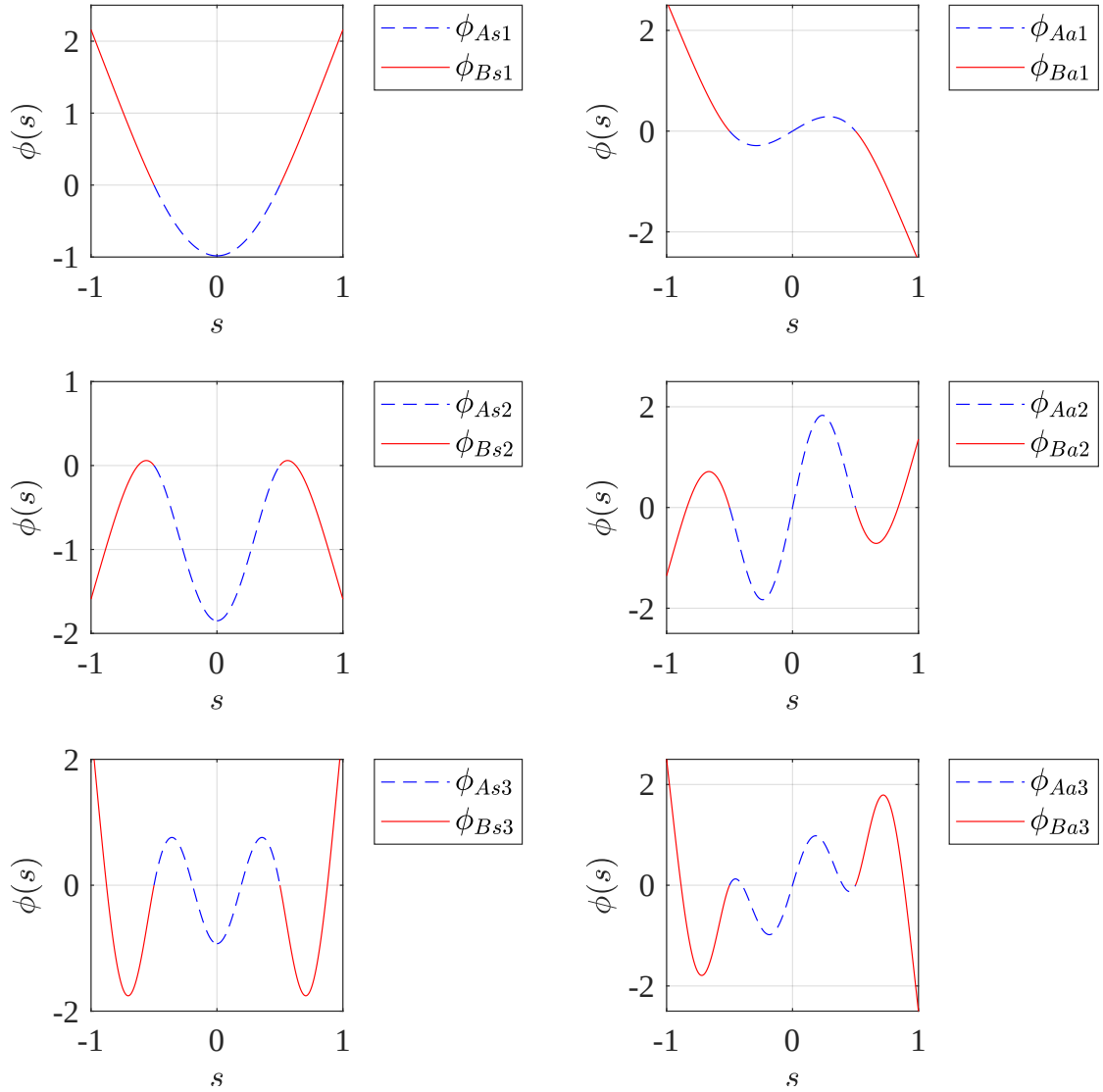


Figure A.21: First three symmetric (left) and anti-symmetric (right) mode shapes, $\kappa = 0.38$.

Appendix B. Reduced model: detailed derivation

Starting with the non-dimensional equations of motion, Eqs. (6) and (7), we substitute $u^A(s,\tau) = \phi_{A1}(s)q_1(\tau) + \phi_{A2}(s)q_2(\tau)$ and $u^B(s,\tau) = \phi_{B1}(s)q_1(\tau) + \phi_{B2}(s)q_2(\tau)$ over the corresponding domains to obtain:

$$
\begin{aligned}
&\left\{\ddot{q}_1 + \tilde{\zeta} z_1^4 \dot{q}_1 + z_1^4 q_1\right\}\phi_{A1} + \left\{\ddot{q}_2 + \tilde{\zeta} z_2^4 \dot{q}_2 + z_2^4 q_2\right\}\phi_{A2} + \tilde{\beta}\left[\phi''_{A1} q_1 + \phi''_{A2} q_2\right] \\
&- \tilde{\alpha}\left[\left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi'_{A1})^2 ds\right\}\phi''_{A1} q_1^3 + \left(\left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi'_{A1})^2 ds\right\}\phi''_{A2} + 2\cancelto{0}{\left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} \phi'_{A1}\phi'_{A2} ds\right\}}\phi''_{A1}\right) q_1^2 q_2\right. \\
&\left.\left(\left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{2}(\phi'_{A2})^2 ds\right\}\phi''_{A1} + 2\cancelto{0}{\left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} \phi'_{A1}\phi'_{A2} ds\right\}}\phi''_{A2}\right) q_1 q_2^2 + \left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi'_{A2})^2 ds\right\}\phi''_{A2} q_2^3\right] = F(q_1, q_2, \tau) \\
\Longrightarrow\ &\left\{\ddot{q}_1 + \tilde{\zeta} z_1^4 \dot{q}_1 + z_1^4 q_1\right\}\phi_{A1} + \left\{\ddot{q}_2 + \tilde{\zeta} z_2^4 \dot{q}_2 + z_2^4 q_2\right\}\phi_{A2} + \tilde{\beta}\left[\phi''_{A1} q_1 + \phi''_{A2} q_2\right] - F(q_1, q_2, \tau, s) = \\
&\tilde{\alpha}\left[\left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi'_{A1})^2 ds\right\}\phi''_{A1} q_1^3 + \left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi'_{A1})^2 ds\right\}\phi''_{A2} q_1^2 q_2 + \left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi'_{A2})^2 ds\right\}\phi''_{A1} q_1 q_2^2 + \left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi'_{A2})^2 ds\right\}\phi''_{A2} q_2^3\right] \quad \text{(B.1)}
\end{aligned}
$$

$$
\left\{\ddot{q}_1 + \tilde{\zeta} z_1^4 \dot{q}_1 + z_1^4 q_1\right\}\phi_{B1} + \left\{\ddot{q}_2 + \tilde{\zeta} z_2^4 \dot{q}_2 + z_2^4 q_2\right\}\phi_{B2} = G(q_1, q_2, \tau, s) \quad \text{(B.2)}
$$

where the cancelled terms result from integrating products of even and odd functions over a symmetric interval, noting that derivation of (hyperbolic) trigonometric functions inverses their parity. The Galerkin method is completed by multiplying Eqs. (B.1) and (B.2), respectively, by ϕ_{Aj} and ϕ_{Bj}, where $j = 1, 2$, and integrating over their corresponding intervals. For $j = 1$, the terms in the above equations proportional to ϕ_{x2} or ϕ''_{x2}, with $x = A, B$, vanish due to orthogonality, leading to:

$$
\begin{aligned}
&\left\{\ddot{q}_1 + \tilde{\zeta} z_1^4 \dot{q}_1 + z_1^4 q_1\right\}\left[\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi_{A1})^2 ds + \int_{-1}^{-\frac{1}{2}} (\phi_{B1})^2 ds + \int_{\frac{1}{2}}^{1} (\phi_{B1})^2 ds\right] + \tilde{\beta}\left[\int_{-\frac{1}{2}}^{\frac{1}{2}} \phi_{A1}\phi''_{A1} ds\right] q_1 \\
&- \tilde{\alpha}\left[\left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi'_{A1})^2 ds\right\}\left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} \phi_{A1}\phi''_{A1} ds\right\} q_1^3 + \left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi'_{A2})^2 ds\right\}\left\{\int_{-\frac{1}{2}}^{\frac{1}{2}} \phi_{A1}\phi''_{A1} ds\right\} q_1 q_2^2\right] \\
&= \phi_{A1}(0) F(q_1, q_2, \tau, 0) + \phi_{B1}(-1) G(q_1, q_2, \tau, -1) + \phi_{B1}(1) G(q_1, q_2, \tau, 1)
\end{aligned}
$$

This expression motivates the use of:

$$
\int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi_{A1})^2 ds + \int_{-1}^{-\frac{1}{2}} (\phi_{B1})^2 ds + \int_{\frac{1}{2}}^{1} (\phi_{B1})^2 ds = 1
$$

as a normalization condition. Moreover, we note that $\phi_{B1}(-1) = \phi_{B1}(1)$. Further simplification is possible by realizing that:

$$
I_j = \int_{-\frac{1}{2}}^{\frac{1}{2}} \phi_{Aj}\phi''_{Aj} ds = \cancelto{0}{\left[\phi_{Aj}\phi'_{Aj}\right]_{-\frac{1}{2}}^{\frac{1}{2}}} - \int_{-\frac{1}{2}}^{\frac{1}{2}} (\phi'_{Aj})^2 ds \quad \text{(B.3)}
$$

due to compatibility conditions, Eq. (10), for all modes. Thus:

$$\ddot{q}_1 + \tilde{\zeta} z_1^4 \dot{q}_1 + \left(z_1^4 + \tilde{\beta} I_1\right) q_1 + \tilde{\alpha}\left[I_1^2 q_1^3 + I_1 I_2 q_1 q_2^2\right] = \phi_{A1}(0) F(q_1, q_2, \tau, 0) + \phi_{B1}(1)\left[G(q_1, q_2, \tau, 1) + G(q_1, q_2, \tau, -1)\right]$$

In the same way, and considering $\phi_{B2}(-1) = -\phi_{B2}(1)$, $\phi_{A2}(0) = 0$, we obtain the second-mode equation of motion:

$$\ddot{q}_2 + \tilde{\zeta} z_2^4 \dot{q}_2 + \left(z_2^4 + \tilde{\beta} I_2\right) q_2 + \tilde{\alpha}\left[I_1 I_2 q_1^2 q_2 + I_2^2 q_2^3\right] = \phi_{B2}(1)\left[G(q_1, q_2, \tau, 1) - G(q_1, q_2, \tau, -1)\right]$$

The form given in Eqs. (13) and (14) is thus obtained by letting:

$$\tilde{a}_{11} = I_1^2 \qquad \tilde{a}_{12} = \tilde{a}_{21} = I_1 I_2 \qquad \tilde{a}_{22} = I_2^2 \tag{B.4}$$

Appendix C. Nondimensional parameters

Nonlinear curvature coefficients:

$$\alpha_{11} = \tilde{\alpha}\tilde{a}_{11}\left(\frac{\gamma}{\hat{\omega}_1 \phi_1^B(1)}\right)^2 \qquad \alpha_{12} = \tilde{\alpha}\tilde{a}_{12}\left(\frac{\gamma}{\hat{\omega}_1 \phi_2^B(1)}\right)^2$$
$$\alpha_{21} = \tilde{\alpha}\tilde{a}_{21}\left(\frac{\gamma}{\hat{\omega}_1 \phi_1^B(1)}\right)^2 \qquad \alpha_{22} = \tilde{\alpha}\tilde{a}_{22}\left(\frac{\gamma}{\hat{\omega}_1 \phi_2^B(1)}\right)^2$$

Hence, for a given α_{11}, the remaining coefficients can be computed as:

$$\alpha_{12} = \alpha_{11}\frac{I_2}{I_1}\left(\frac{\phi_1^B(1)}{\phi_2^B(1)}\right)^2 \qquad \alpha_{21} = \alpha_{11}\frac{I_2}{I_1} \qquad \alpha_{22} = \alpha_{11}\left(\frac{I_2 \phi_1^B(1)}{I_1 \phi_2^B(1)}\right)^2 \tag{C.1}$$

Nonlinear damping coefficients:

$$\eta_{11} = \tilde{\eta}\frac{\tilde{a}_{11}}{\hat{\omega}_1}\left(\frac{\gamma}{\phi_1^B(1)}\right)^2 \qquad \eta_{12} = \tilde{\eta}\frac{\tilde{a}_{12}}{\hat{\omega}_1}\left(\frac{\gamma}{\phi_2^B(1)}\right)^2$$
$$\eta_{21} = \tilde{\eta}\frac{\tilde{a}_{21}}{\hat{\omega}_1}\left(\frac{\gamma}{\phi_1^B(1)}\right)^2 \qquad \eta_{22} = \tilde{\eta}\frac{\tilde{a}_{22}}{\hat{\omega}_1}\left(\frac{\gamma}{\phi_2^B(1)}\right)^2$$

Hence, for a given η_{11}, the remaining coefficients can be computed as:

$$\eta_{12} = \eta_{11}\frac{I_2}{I_1}\left(\frac{\phi_1^B(1)}{\phi_2^B(1)}\right)^2 \qquad \eta_{21} = \eta_{11}\frac{I_2}{I_1} \qquad \eta_{22} = \eta_{11}\left(\frac{I_2 \phi_1^B(1)}{I_1 \phi_2^B(1)}\right)^2 \tag{C.2}$$

Excitation coefficients:

$$\mu = -\tilde{\mu}\frac{\left[\phi_1^B(1)\right]\left[\phi_1^A(0)\right]}{\hat{\omega}_1^2 \gamma} \qquad d_0 = \frac{\gamma\left[\left(\phi_2^A\right)'(0)\right]}{\left[\phi_2^B(1)\right]} \qquad \epsilon_{1,2} = \frac{\tilde{\epsilon}\left[\phi_{1,2}^B(1)\right]^2}{\hat{\omega}_1^2 \gamma^3} \tag{C.3}$$

Appendix D. Overview of the HBM-AFT method

Appendix D.1. Cycle computation

We consider the dynamics of the n degrees of freedom $\mathbf{q}(t) \in \mathbb{R}^n$, given by the equations of motion:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} + \mathbf{f}_{\mathrm{NL}}(\mathbf{q}, \dot{\mathbf{q}}, t) = \mathbf{p} \tag{D.1}$$

where the matrices $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ represent the system's linear inertia, damping and stiffness, respectively, $\mathbf{p}$ accounts for external forcing, and all nonlinear effects are grouped in vector $\mathbf{f}_{\mathrm{NL}}$. The latter can be a function of time if, for instance, parametric excitation is considered. Moreover, arbitrary nonlinearities are allowed so long as they are smooth functions of $\mathbf{q}$ and/or $\dot{\mathbf{q}}$, with an extension to $\ddot{\mathbf{q}}$-dependent forces being straightforward. We seek periodic solutions of Eq.(D.1) with base period $T = 2\pi/\omega$, where the frequency is imposed by either external ($\mathbf{p}(t)$) or parametric ($\mathbf{f}_{\mathrm{NL}}$) excitation. Thus, $\mathbf{q}(t + T) = \mathbf{q}(t)$, and any particular DOF may be uniquely expressed as a Fourier series which we truncate at H harmonics:

$$q_j(t) = Q_j^0 + \sum_{k=1}^{H} \left[Q_j^{ck} \cos(k\omega t) + Q_j^{sk} \sin(k\omega t) \right] = \Gamma_H(\omega t)\mathbf{Q}_j \implies \mathbf{q}(t) = (\Gamma_H(\omega t) \otimes \mathbf{I}_n)\,\mathbf{Q} \tag{D.2}$$

in which vector $\mathbf{Q}_j$ contains the Fourier coefficients of the j-th DOF and $\Gamma_H(\omega t)$ groups the trigonometric basis functions. We note that m-th order subharmonic components are readily included in this expansion by simply replacing ω by ω/m, $m \in \mathbb{N}$, in Γ_H. For $n > 1$, the Kronecker product allows us to write succinctly: $\mathbf{q}(t) = (\Gamma_H(\omega t) \otimes \mathbf{I}_n)\,\mathbf{Q}$, where $\mathbf{Q}$ groups the Fourier coefficients of all DOFs. As only Γ_H is time-dependent, velocities and accelerations are also proportional to $\mathbf{Q}$, as follows: $\dot{\mathbf{q}} = \omega(\Gamma_H(\omega t)\nabla \otimes \mathbf{I}_n)\mathbf{Q}$, $\ddot{\mathbf{q}} = \omega^2(\Gamma_H(\omega t)\nabla^2 \otimes \mathbf{I}_n)\mathbf{Q}$, where ∇ is a linear permutation operator. The periodicity assumption requires $\mathbf{p}$ and $\mathbf{f}_{\mathrm{NL}}$ to be T-periodic as well, so we expand them similarly and associate them with Fourier coefficients $\mathbf{P}$ and $\mathbf{F}_{\mathrm{NL}}$, respectively. All of these expressions are inserted into Eq. (D.1), to which we then apply a Galerkin projection by scalar product with $(\Gamma_H^{-1}(\omega t) \otimes \mathbf{I}_n$. This yields the system of nonlinear algebraic equations:

$$\mathbf{R}(\mathbf{Q}) = \mathbf{Z}\mathbf{Q} + \mathbf{F}_{\mathrm{NL}}(\mathbf{Q}) - \mathbf{P} = \mathbf{0} \tag{D.3}$$

which can be solved numerically through any root-finding method. The dynamic stiffness matrix $\mathbf{Z}$ and the external forcing vector $\mathbf{P}$ are independent of $\mathbf{Q}$ and easy to compute. On the other hand, in all but the simplest cases, a closed-form expression for the Fourier coefficients $\mathbf{F}_{\mathrm{NL}}$ is unavailable. Nevertheless, given a set of coefficients $\mathbf{Q}$, their corresponding time series -discretised over one period of motion- can be assembled by inverse Fourier transform, i.e. by replacing $\Gamma_H(\omega t)$ in Eq. (D.2) by a call to a suitable subroutine. This allows us to compute $\mathbf{f}_{\mathbf{NL}}$ from a known closed-form expression, and then to extract its Fourier coefficients through Fourier transform. Symbolically: $\mathbf{F}_{\mathrm{NL}} = \left(\Gamma_H^{-1} \otimes \mathbf{I}_n\right)\mathbf{f}_{\mathrm{NL}}$, although in practice we use a specialised subroutine implementing a 'fast' version of this operation in order to greatly enhance performance. Hence, using the Newton-Raphson method for the sake of argument, the approximation at the $(j + 1)$-th iteration is given by:

$$\mathbf{Q}^{(j+1)} = \mathbf{Q}^{(j)} - \left[\mathbf{R}_{\mathbf{Q}}^{(j)}\right]^{-1} \mathbf{R}^{(j)} \tag{D.4}$$

where the superscripts indicate evaluation at $\mathbf{Q}^{(j)}$, and the Jacobian is computed explicitly as:

$$\mathbf{R_Q} = \mathbf{Z} + \frac{d\mathbf{F}_{\mathrm{NL}}}{d\mathbf{Q}} = \mathbf{Z} + \left(\Gamma^{-1} \otimes \mathbf{I}_n\right)\left\{\frac{\partial \mathbf{f}_{\mathrm{NL}}}{\partial \mathbf{q}}\left(\Gamma \otimes \mathbf{I}_n\right) + \omega \frac{\partial \mathbf{f}_{\mathrm{NL}}}{\partial \dot{\mathbf{q}}}\left(\Gamma\nabla \otimes \mathbf{I}_n\right)\right\} \tag{D.5}$$

In the above expression, we once again take advantage of the availability of closed-form expressions in time domain for efficient computation of derivatives. Iterations are performed until the norm of the residual vector, $\mathbf{R}$, fall below a given tolerance. It is important to note that the quality of a solution obtained in this manner strongly depends on the chosen number of harmonics, H. While an infinite number of terms is rigorously required for any series solution to be exact, the magnitudes of harmonic coefficients higher than the fundamental decay rapidly and thus become negligible beyond some threshold value H_c. A convergence analysis, using a criterion such as the norm of $\mathbf{F}_{\mathrm{NL}}$ for varying H, should thus be used to determine an optimal number of harmonics for a given problem.

Appendix D.2. Stability analysis: Hill's method

Given a periodic solution to Eq. (D.1), denoted $\mathbf{q}_0(t)$, we assess its local stability by considering a slightly perturbed state, $\mathbf{q}(t) = \mathbf{q}_0(t) + \mathbf{s}(t)$, where $\|\mathbf{s}(t)\| << 1$. A first-order Taylor expansion then yields the linearised equations about the periodic solution, as:

$$\mathbf{M}\ddot{\mathbf{s}} + \left[\mathbf{C} + \left.\frac{\partial \mathbf{f}_{\mathrm{NL}}}{\partial \dot{\mathbf{q}}}\right|_{\mathbf{q}_0}\right]\dot{\mathbf{s}} + \left[\mathbf{K} + \left.\frac{\partial \mathbf{f}_{\mathrm{NL}}}{\partial \mathbf{q}}\right|_{\mathbf{q}_0}\right]\mathbf{s} = \mathbf{0} \tag{D.6}$$

Eq. (D.6) is a linear, homogeneous system of equations with periodic coefficients. This is the framework of Floquet theory, from which it can be shown that there exist $2n$ linearly independent solutions of the form $\mathbf{s}_j(t) = e^{\lambda_j t}\phi(t)_j$, $j = 1, ..., 2n$, where $\phi_j(t + T) = \phi_j(t)$. The Floquet exponents λ_j govern the stability of cycle $\mathbf{q}_0$: the existence of any one of them with positive real part translates into exponentially amplified perturbations over time, characteristic of an unstable cycle. Conversely, $\Re\left(\lambda_j\right) < 0\ \forall j$ is true for a stable cycle. In the context of the HBM, it is convenient to introduce the form of $\mathbf{s}(t)$ into Eq. (D.6), expand the periodic function into a Fourier series: $\phi_j(t) = \left(\Gamma_H \otimes \mathbf{I}_n\right)\Phi_j$, and perform a Galerkin projection. This leads to a quadratic eigenvalue problem for the Floquet exponents:

$$\left[\mathbf{R_Q} + \lambda_j \mathbf{D}_1 + \lambda_j^2 \mathbf{D}_2\right]\Phi_j = \mathbf{0} \tag{D.7}$$

where the matrices $\mathbf{D}_1$ and $\mathbf{D}_2$ are simple functions of $\mathbf{M}$, $\mathbf{C}$ and ω, and the Jacobian $\mathbf{R_Q}$ is already available from the final iteration of the root-finding procedure. We note that the size of Eq. (D.7) is $2n(2H + 1)$, from which only $2n$ are the actual Floquet exponents while the remaining eigenvalues are a mathematical artifice due to the truncation of an infinite series. This is not an issue in practice, however, as it can been formally proven [33] that the $2n$ solutions with smallest imaginary part in magnitude converge to the correct Floquet exponents with increasing H. Moreover, when convergence is achieved, the condition: $|\Im(\lambda_j)| \leq \omega/2$ is verified by all Floquet exponents. Monitoring these eigenvalues along continuation allows us to detect bifurcations, localize them, and track them with respect to the system's parameters.

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